RIME Computational Companion Archive

Versioned Reproducibility Data, Computational Observations, Open Problems, and Historical Records

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Archive boundary. This archive is optional human-readable research companion material. Papers I, II, and III are mathematically self-contained and do not rely on it as a premise, definition source, executable certificate, or claim authority. It is not a paper, theorem source, semantic authority, or prerequisite for those papers. Mathematical claims must be cited from the corresponding papers. The archive preserves reproducibility data, computational observations, open conjectures, and versioned historical records with explicit status labels. Executable certificates are controlled by their declared scripts and structured artifacts; corrections affecting interpretation are recorded in `HISTORY.md`.

Independent Paper Navigation

The entries below are navigation records, not a dependency chain. Papers I, II, and III have independent Zenodo records.

Paper	Independent title	Repository	Separate Zenodo DOI
I	*Spectral Sector Decomposition in the Rubik's Cube Representation: Block Spectral Structure and a Conditional Rationality Criterion*	PDF, source, BibTeX `paper1`	10.5281/zenodo.21571403
II	*Noncommutative Transport Topology in the Rubik's Cube Representation: Sector Non-Invariance, Direct Support, and Transport Channels*	PDF, source, BibTeX `paper2`	10.5281/zenodo.21581072
III	*Support-Graph Reachability and Matrix-Composition Obstructions: Image–Kernel Mismatch with a Rubik-Cube Case Study*	PDF, source, BibTeX `paper3`	10.5281/zenodo.21583070

The immutable combined-release DOI 10.5281/zenodo.21108197 remains historical provenance. It does not define the current Paper I–III architecture and must not be reused as the CCS v2 DOI. That historical combined package is a provenance record only.

Paper II v2 routing. The Paper II theorem spine is the exact Transport–Non-Invariance Identity and direct block-locality theorem. The nine-sector registration, ten-edge graph, Type I/II labels, and EP algebra census are computational certificates. Block-level $\mathrm{Supp}_{\mathrm{nc}}$ is the family-level maximum over all three per-axis QT commutator pairs and is only a candidate localizer: its 15 overlap pairs contain the nine Type I labelled edges and six nonedges. It is not a sufficient transport criterion. Generator-family field tables, S3 negative controls, full EP algebra tables, and auxiliary figures remain CCS material and are not manuscript premises.

Reproducibility Architecture

The repository uses four separated layers:

Layer	Name	Files	Role	Rule
1	Papers	`papers/paper*/`	Claim	Self-contained theorem, certificate, observation, and research-program boundaries.
2	Executable artifacts	`experiments/paper*/`, `results/`	Certify	Declared scripts, parameters, hashes, residuals, and structured outputs.
3	CCS v2	`ccs/canonical_specification.md`	Review	Human-readable extended data, observations, open questions, and selected history.
4	Private raw archive	not distributed	Preserve	Giant matrices, abandoned runs, and exploratory provenance.

CCS v2 is not a copy of the raw archive and does not replace executable artifacts. It is a curated review layer. The papers state their claims, the declared scripts and structured artifacts certify finite computations, and `HISTORY.md` records corrections that affect interpretation.

Navigation and Citation Boundary

Papers I–III do not cite CCS v2 as a scholarly or mathematical authority. Internal part, table, and figure labels below are navigation aids for this archive only. Paper III numerical claims are defined by its own manuscript and matrix certificate.

Internal part references. Archive-local links use Roman numeral prefixes:

Prefix	Scope	Example
`CCS Part 0`	Global Reference Map (notation tables, terminology)	`(CCS Part 0)`
`CCS Part 0.5`	Canonical API Surface	`(CCS Part 0.5)`
`CCS-I`	Part I — Core Numerical Structures (§1, §2)	`(CCS-I §2.1)`
`CCS-II`	Part II — Extended Computational Observations (§II.1–II.4)	`(CCS-II §II.4)`
`CCS-III`	Historical derivation index; excluded from the release PDF	source provenance only
`CCS Appendix X`	Appendices A–F	`(CCS Appendix F)`

Section numbering. Part I uses §1.x (spectral objects) and §2.x (numerical data). Part II uses §II.1–II.4 for extended observations. The historical Part III numbering is retained in source provenance only. Appendix subsections use letter prefixes (§A.1, §B.1, §E.1, §F.1).

Tables and figures. `(CCS Table C3)`, `(CCS Fig. C1)` — figures are captioned where they appear. Historical images are repository provenance and are not separately indexed in this release.

The Terminology Convention at the end of Part 0 defines the four canonical terms: QT/HT joint-spectral sector, hybrid sector, transport-active, canonical sectorization.

Archive Status Tags

Every retained item must carry one of the following statuses in context:

Status	Meaning
Theorem / exact derivation	Finite mathematical statement proved under explicit hypotheses; the independent paper remains the preferred citation.
Computational certificate	Declared realization, dtype, tolerance, algorithm, artifact, and reproducible script are available.
Computational observation	Finite pattern or numerical recognition without a full promotion certificate.
Research program	Conjecture, proposed hierarchy, genericity question, or future experiment.
Historical / withdrawn	Provenance only; excluded from the PDF or displayed with an explicit historical warning.

Box Conventions

Callout boxes separate exact statements, registered data, warnings, and provisional findings. Their labels describe archive status; they do not make CCS v2 an independent claim authority:

Box	Style	Purpose
Theorem / Lemma / Corollary / Definition	Blockquote > with bold label	Formal mathematical statement. Proofs appear inside the box, set off with *Proof.* or *Proof sketch.*
Registered	Blockquote > with **Registered.** label	Finite data tied to a declared computational realization.
Warning	Blockquote > with **Warning.** label	Important constraint, pitfall, or normative requirement (SHALL/MUST). Non-negotiable.
Exploratory	Blockquote > with **Exploratory.** label	Observation, conjecture, or research-program item subject to revision.

1 Executive Guide

This guide provides the shortest reliable reading path through the archive. The subsequent sections preserve tables, implementation context, figures, failed candidates, and research history. They do not enlarge the claims of the independent papers.

Current Finite Records

Record	Current status	Owning source
Six displayed averaging layers with dimension census `(20,2,39,26,106,35)`	Computational observation; exact and conditional statements are separated in Paper I	Paper I and `experiments/paper1/validation/`
Nine registered QT/HT joint-spectral sectors	Numerical registration on the declared complex128 realization	Paper II and `experiments/paper2/validation/`
Symmetric ten-edge direct graph with degree sequence `(0,2,2,2,2,5,3,1,3)`	Computational certificate	Paper II
Fifteen noncommutative-support candidates: nine Type I edges and six nonedges	Computational certificate; the localizer is not sufficient	Paper II
EP algebra census: four $M_2(\mathbb{C})$ components and four scalar components	Computational certificate supported by the finite-dimensional unital $*$-algebra argument	Paper II
Five support-graph paths whose evaluated projected products are machine-zero	Computational certificate with image–kernel/block obstruction interpretation	Paper III

These five declared two-step support-graph paths form the finite Paper III composition-obstruction audit.

Promotion Boundaries

- Numerical QT/HT commutation and clustering do not prove an exact labelled joint spectral resolution.
- Numerical recognition of rational or quadratic values does not determine an exact spectral field.
- A direct-support path does not guarantee a nonzero routed matrix product.
- A routed product does not automatically determine a full word, commutator, or Lie depth.
- Ambient incidence codimension is a benchmark, not the codimension of a representation-derived pullback.

Research Seeds Preserved Here

The archive retains three concrete promotion targets: exact QH algebra registration; exact characteristic/minimal-polynomial certificates for generator-family arithmetic contrasts; and structured pullback geometry for representation-derived incidence. Earlier T7, commutant-restriction, completion, search, and spectral-triple narratives are historical records only.

Part 0 — Global Reference Map

Purpose. Human-readable lookup for registered Paper I–II numerical objects, implementation locations, and shared Rubik notation. The independent papers define their own mathematical objects and claims locally.

How to use. For any symbol or concept name, use column 3 to locate the paper definition or the archive record and column 4 to identify its scope. Paper definitions control mathematical meaning; archive entries provide data and provenance only.

Registered default. Unless otherwise stated, the archived nine-sector decomposition is the numerical joint-spectral registration associated with

$$\mathcal{B}_{\mathrm{QH}} = \mathrm{alg}(A_{18}, \mathrm{QT}_{\mathrm{all}}, \mathrm{HT}_{\mathrm{all}}) = \mathrm{alg}(\mathrm{QT}_{\mathrm{all}}, \mathrm{HT}_{\mathrm{all}}),$$

where $A_{18} = (2/3)\mathrm{QT}_{\mathrm{all}} + (1/3)\mathrm{HT}_{\mathrm{all}}$. The displayed algebraic interpretation is conditional on exact commuting-Hermitian registration. Sectorizations involving auxiliary block projectors (e.g. P_{nat}) are separate declared realizations.

Layer A — Static Spectral Structure (Paper I)

Core object: $A = \frac{1}{|S|}\sum_{g\in S}\rho(g)$

Symbol	Concept	First Defined	Used In
$\rho : G \to \mathrm{GL}(228, \mathbb{C})$	Rubik's cube representation	Paper I §2	I, II, III
$V = V_{\mathrm{cp}} \oplus V_{\mathrm{ep}} \oplus V_{\mathrm{co}} \oplus V_{\mathrm{eo}}$	Block decomposition $(64 + 144 + 8 + 12 = 228)$	Paper I §3.4	I, II, III
cp, ep, co, eo	Four invariant blocks	Paper I §2	I, II, III
$A = \frac{1}{|S|}\sum_{s\in S}\rho(s)$	Averaging operator	Paper I §2	I, II, III
V_λ, $\lambda = 1 - k/9$	Canonical layers (6), eigenvalue form	Paper I §3	I, II, III
$k \in \{0, 1, 2, 3, 4, 6\}$	Admissible k-set (6 values, $k = 5$ vacant)	Paper I §3, §7.3	I, II, III
P_λ^A	Orthogonal projector onto the A-eigenspace V_λ^A	Paper I §3.1	I, II
$\mathcal{K}(A) = \bigcup_B \mathcal{K}_B$	Blockwise k-set union formula	Paper I §7.3	I
$\chi_\lambda(s) = \mathrm{Tr}(P_\lambda \rho(s))$	Eigenspace trace	Paper I §3.1	I
$\omega + \omega^2 + 1 = 0$	$\mathbb{Z}_3$ phase cancellation	Paper I §4.1, §7.1	I
$h_i = \frac{1}{2}(\rho(g_i) + \rho(g_i^{-1}))$	Per-generator Hermitian average	Paper I §7.2	I
$V = \bigoplus_\mu V^{(\mu)}$, C_μ	Ambient G-isotypic decomposition and its central projectors; distinct from the A-spectral decomposition	Paper I App B	I
$\mathrm{Tr}(P_\lambda^A C_\mu P_\lambda^A)$	Ambient-isotypic overlap mass; not a subrepresentation multiplicity unless the projectors commute	Paper I App B	I
O, U_R	Registered orientation-preserving cubic rotation action commuting numerically with A; distinct from transport by G	Paper I §4, App B	I
$\dim \mathrm{End}_G(V) = 610$	Candidate ambient-commutant dimension; unpromoted pending exact certificate	CCS legacy §2.8	provenance only

Layer B — Discrete Transport Structure (Paper II)

Core object: $K_{\beta\alpha}^S = \max_{s\in S} \|Q_\beta \rho(s) Q_\alpha\|_F$

Symbol	Concept	First Defined	Used In
$S1$–$S9$	Nine numerically registered QT/HT joint-spectral sectors	Paper II §2	II, III
$\mathcal{B}_{\mathrm{QH}} = \mathrm{alg}(A, \mathrm{QT}_{\mathrm{all}}, \mathrm{HT}_{\mathrm{all}})$	Conditional exact algebra associated with the numerically commuting QT/HT registration; not identified with the full group commutant	Paper II §2	II, III
$\mathrm{QT}_{\mathrm{all}}$, $\mathrm{HT}_{\mathrm{all}}$	Quarter-turn / half-turn total averages	Paper II §2	II, III
Q_α	QT/HT joint-spectral sector projector; not assumed G-invariant	Paper II §2	II, III
$K^S_{\beta\alpha} = \max_{s\in S} \lVert Q_\beta \rho(s) Q_\alpha \rVert_F$	Direct generator-transport norm	Paper II §3.1	II, III
$\mathrm{Supp}_{\mathrm{nc}}(\alpha)$	Thresholded family-level block localizer using the maximum over all three per-axis QT commutator pairs; Type I labels require overlap by definition, but overlap is not sufficient	Paper II §4	II
$\lVert[\mathrm{QT}^0, \mathrm{QT}^1]\rVert_b$	Per-block QT commutator norm	Paper II §4.2	II, III
QT^a $(a \in \{0, 1, 2\})$	Per-axis quarter-turn averaging operators	Paper II App A	II, III
Type I / Type II transport	Post-certification labels for the nine shared-noncommutative-support edges and the one CP exception; not universal sufficient criteria	Paper II §4	II
CP permutation channel	Registered S8↔S9 Type II edge ($K = 2.83$)	Paper II §4	II
$A_{\mathrm{EP}} \cong M_2(\mathbb{C})^4 \oplus M_1(\mathbb{C})^4$	Computational EP block algebra census (20-dimensional with 8-dimensional center)	Paper II §4	II
QH refinement boundary	Conditional minimality only inside the declared commuting QH algebra; no global maximal-refinement theorem	Paper II §5	II
Hub pattern	Sparse ten-edge graph with unique degree-five hub S6; the graph is not a star	Paper II §3.4	II, III
S1 isolation	Machine-zero off-diagonal direct transport in the canonical audit; exact G-invariance remains unproved	Paper II §4	II
Generator-family comparison	Extended computational census retained in CCS; not part of the Paper II theorem spine	CCS Part II	CCS only
$T_{\beta\alpha}(g) = Q_\beta \rho(g) Q_\alpha$	Generator transport block from source α to target β	Paper II §3.1	II, III
$\sum_{\beta\neq\alpha} \lVert T_{\beta\alpha}(g)\rVert_F^2 = \frac{1}{2}\lVert[\rho(g), Q_\alpha]\rVert_F^2$	Off-diagonal transport–non-invariance identity	Paper II Prop. 3.5	II, III

External Independent Paper III

The independent Paper III defines its support graph, projected composition operators, image–kernel obstruction, local promotion criteria, and Rubik matrix certificate in its own manuscript. The CCS does not define, number, or certify that theorem spine.

Prototypes, Controls, and Cross-Cutting

Symbol	Concept	First Defined	Used In
S_3 nat⊕reg (9-dim)	Archived first-version sector-invariance control; not matrix-composition evidence	excluded source provenance	provenance only
S_3 reg⊕reg (12-dim)	Archived first-version sector-invariance control; not matrix-composition evidence	excluded source provenance	provenance only
N=2 pocket cube (72-dim)	Archived first-version graph/kappa control; not matrix-composition evidence	CCS legacy control	provenance only
Archived S1–S6 summaries	First-version empirical summaries; current status is determined item by item	CCS legacy §II.5	archive only
Cross-paper comparison path	Spectral sectors → direct support graph → projected composition audit	Independent Papers I–III	comparison only
Canonical layer keys	$\lambda = 1 - k/9$: $[1, 8/9, 7/9, 2/3, 5/9, 1/3]$	CCS Part 0.5	I, II, III
$m = \lvert S\rvert/2$	Effective generator count	Paper I §7.3	I
Face-symmetric / symmetry-broken	Generator family classification	Paper I §7.4	I, II
Registered arithmetic contrast	Historical generator-family scans with values numerically recognized in $\mathbb{Q}$ or $\mathbb{Q}(\sqrt{5})$	CCS Part II	archive only

Terminology Convention

Term	Definition
QT/HT joint-spectral sector	One of the nine numerical joint-spectral clusters registered from the declared QT/HT averages. Under exact commuting-Hermitian registration, these become the joint spectral subspaces associated with primitive spectral idempotents of the generated commutative algebra.
hybrid sector	A QT/HT joint-spectral sector whose projector has nonzero support on more than one block. There are 6 hybrid sectors: S1 (cp+ep), S3 (ep+eo), S4 (ep+co), S6 (ep+eo), S7 (cp+ep+co+eo), S9 (cp+co). S7 is the unique all-block hybrid spanning all four blocks.
transport-active	A sector pair (α, β) is transport-active if $K_{\alpha\beta} > 0$ (non-zero one-step transport). All 10 direct edges are block-preserving.
registered QH sectorization	The nine-sector numerical decomposition obtained from the declared QT/HT pair. Exact commutation is a hypothesis for the corresponding algebraic joint-resolution statement. It is not a full-G commutant decomposition.

Geometric & move conventions (coordinate system, cubie ordering, generator encoding, action direction, block decomposition, numerical tolerances) are maintained in `docs/conventions.md`.

Part 0.5 — Registered API Surface

Purpose. Record the current mapping between mathematical notation, computational interfaces, and archived numerical outputs. This section answers which implementation produced a value; it does not make that implementation a mathematical definition.

Scope. Functions listed here are the registered v2 implementation entry points. Alternative implementations may be used when their conventions, parameters, and comparison residuals are declared.

Dependencies. `rime.cubieoperator.CubieSpectralOperator` (primary), `rime.cubie.CubieMove` (generator enumeration), `rime.spectral_utils` (S_3 negative controls, joint diagonalization helpers).

Outputs. All numerical values in CCS Parts I–II are produced by the functions listed below.

This part records the current paper-to-code mapping; executable scripts and structured artifacts remain the computational certificate layer.

0.5.1 Spectral Objects

Mathematical object	Canonical API	Returns	Stability
Layer eigenvalues	`CubieSpectralOperator().layer_keys`	`list[float]` — 6 canonical λ, descending (property, not method)	**A**
Layer dimension	`.layer_dimension(lam)`	`int`	**A**
Layer projector	`.layer_projector(lam)`	`ndarray` ($228{\times}228$)	**A**
Closest layer	`.closest_layer(lam)`	`float` — canonical λ key	**A**
Sector decomposition	`.center_decomposition()`	`dict` with `n_sectors`, `projectors`, `sectors`	**A**
A_18 operator	`.A` (property)	`ndarray` ($228{\times}228$)	**A**

0.5.2 Transport and Accessibility

Mathematical object	Canonical API	Returns	Stability
Direct-support matrix plus legacy arrays	`.transport_kappa(projectors, compute_kappa1=True)`	`tuple[K, kappa0, kappa1]`; only K is the current direct-support object	**B/C**
First-version κ array at depth d	`.kappa_depth(d)`	archived principal-log diagnostic; not current exact Lie depth	**C**
Principal-log registration	`.compute_lie_generators()`	`list[ndarray]` — 18 numerically skew-Hermitian matrices for the declared branch	**B/C**
ρ(g) matrices	`.rho_matrices()`	`list[ndarray]` — 18 unitary representation matrices	**A**

0.5.3 Algebraic Structure

Mathematical object	Canonical API	Returns	Stability
Ambient commutant candidate	`.full_commutant_combinatorial()`	numerical/combinatorial candidate basis of dimension 610; exact certificate still required	**C**
Compressed spectral-layer commutants	`.commutant_algebra()`	withdrawn layerwise interpretation; archive compatibility only	**C**
Block projectors	`BLOCK_RANGES` (in `rime.cubie`)	block index slices	**A**
QT/HT per-axis ops	`.build_per_axis_ops()`	$\mathrm{Q}T^0$,$\mathrm{Q}T^1$,QT^2, $\mathrm{H}T^0$,$\mathrm{H}T^1$,HT^2	**A**

0.5.4 S_3 Prototypes

Mathematical object	Canonical API	Returns	Stability
S_3 representations	`rime.spectral_utils.build_s3_*_rep()`	`ndarray` — S_3 irrep matrices	**A**
Joint diagonalization	`rime.spectral_utils.joint_diag_sectors()`	sector projectors	**A**
Two-step projected product maximum	`rime.spectral_utils.max_two_step_composition()`	maximum Frobenius norm and maximizing generator pair	**B**
Graph-only candidate enumeration	`rime.spectral_utils.find_graph_only_two_step_pairs()`	endpoint/intermediate triples requiring matrix audit	**B**
First-version T7 detector	`rime.spectral_utils.find_t7_pairs()`	archived support-level interpretation; not a morphism certificate	**C**

0.5.5 Generator Enumeration

Mathematical object	Canonical API	Returns	Stability
18 face-turn generators	`CubieMove.prim_moves`	`list[CubieMove]`	**A**
Generator weighting	`A_18 = (12 QT_all + 6 HT_all) / 18`	Definitional identity	**A**

Legacy stability key: **A** = invariant under the listed recomputation, permutation, and gauge checks; **B** = stable with fixed parameters and the declared tolerance sweep; **C** = exploratory or withdrawn. These tags describe the archived implementation record and are not the current four-level paper claim status.

Implementation boundary. The listed APIs are the registered implementation paths used to generate this archive. Alternative implementations are allowed when they declare conventions, parameters, tolerances, and comparison residuals. The independent papers and their claim-specific executable artifacts, not this API table, control current claims and certificates.

Part I — Core Numerical Structures

Purpose. Register the numerical objects used by Papers I–II. These data do not replace paper-level proofs and do not govern revised Paper III.

Scope. Operators, eigenspaces, sectors, projectors at 6-layer (A_{18}) and 9-sector (Center) resolution. All canonical tables live in this Part.

Dependencies. The Rubik's cube representation construction (`rime/cubie.py`, `rime/cubieoperator.py`).

Outputs. All objects and numerical values referenced by Parts II–III and the papers.

CCS Fig. C0 omitted. The first-version combined pipeline is not part of the current reproducibility compendium.

1.1 Representation Space

The Rubik's cube group acts on a 228-dimensional complex vector space with four G-invariant blocks:

$$V = V_{\mathrm{cp}} \oplus V_{\mathrm{ep}} \oplus V_{\mathrm{co}} \oplus V_{\mathrm{eo}}, \qquad 64 + 144 + 8 + 12 = 228$$

Table C1 — Block Decomposition.

Block	Dim	Content	Algebra
CP	64	Corner permutation	Q_3 Hamming scheme H(3,2), Bose-Mesner algebra $\cong$ Hecke H($S_2 \wr S_3$, S_3)
EP	144	Edge permutation	Face-incidence adjacency JJ^{T}, noncommutative core
CO	8	Corner orientation ($\mathbb{Z}_3$)	Abelian phase structure
EO	12	Edge orientation ($\mathbb{Z}_2$)	Abelian phase structure

Block order throughout: CP $\to$ EP $\to$ CO $\to$ EO.

1.2 Averaging Operator

$$A = \frac{1}{|S|} \sum_{s \in S} \rho(s)$$

For the canonical 18 face-turn generators ($S = S^{-1}$):

$$A_{18} = (12\,\mathrm{QT}_{\mathrm{all}} + 6\,\mathrm{HT}_{\mathrm{all}})/18$$

where $\mathrm{QT}_{\mathrm{all}} = \sum_{a \in \{x,y,z\}} \mathrm{QT}^a$, $\mathrm{HT}_{\mathrm{all}} = \sum_{a \in \{x,y,z\}} \mathrm{HT}^a$, and

$$\mathrm{QT}^a = \tfrac{1}{2}(\rho(+a) + \rho(-a)), \qquad \mathrm{HT}^a = \rho(2a)$$

A is Hermitian because the declared generator family is inverse closed and the representation is unitary; Paper I states the general result with its hypotheses.

In the declared complex128 realization, the QT/HT averages are registered as numerically commuting. Conditional on exact commutation, the corresponding commutative algebra is

$$Z_{\mathrm{QH}} = \langle A_{18}, \mathrm{QT}_{\mathrm{all}}, \mathrm{HT}_{\mathrm{all}} \rangle = \langle \mathrm{QT}_{\mathrm{all}}, \mathrm{HT}_{\mathrm{all}} \rangle.$$

The nine sectors in §1.4 are numerical joint-spectral clusters. Conditional on exact registration, the six displayed layers are the collision quotient obtained by the linear projection

$$L_{2/3}(q, h) = (2q + h)/3.$$

1.3 Six Canonical Layers

The declared computation registers six eigenspaces of A_{18} against the displayed rational values $\lambda = 1 - k/9$. Conditional on exact QT/HT registration, these are the collision quotients of the nine joint-spectral sectors under $L_{2/3}$.

Table C2 — Six Canonical Layers.

k	λ	dim	Label	Block composition	Layer
0	1	20	V_1	cp(8) + ep(12)	A
1	8/9	2	$V_{8/9}$	eo(2)	A
2	7/9	39	$V_{7/9}$	ep(36) + eo(3)	A
3	2/3	26	$V_{2/3}$	ep(24) + co(2)	A
4	5/9	106	$V_{5/9}$	cp(24) + ep(72) + co(3) + eo(7)	A
6	1/3	35	$V_{1/3}$	cp(32) + co(3)	A

$k = 5$ ($\lambda = 4/9$) is absent from the registered block census; see §1.7 for the exact cp/ep reductions and qualified co/eo audit.

Canonical layer keys: $[1, 8/9, 7/9, 2/3, 5/9, 1/3]$ ($\lambda = 1 - k/9$, $k \in \{0, 1, 2, 3, 4, 6\}$).

1.4 Nine QT/HT Joint-Spectral Sectors

These nine numerical clusters are produced by the declared QT/HT joint diagonalization and registration policy. For commuting Hermitian QT/HT operators, the corresponding projectors are primitive spectral idempotents of the generated commutative algebra. This does not assert a finest orthogonal sectorization in all of $\mathrm{End}(V)$.

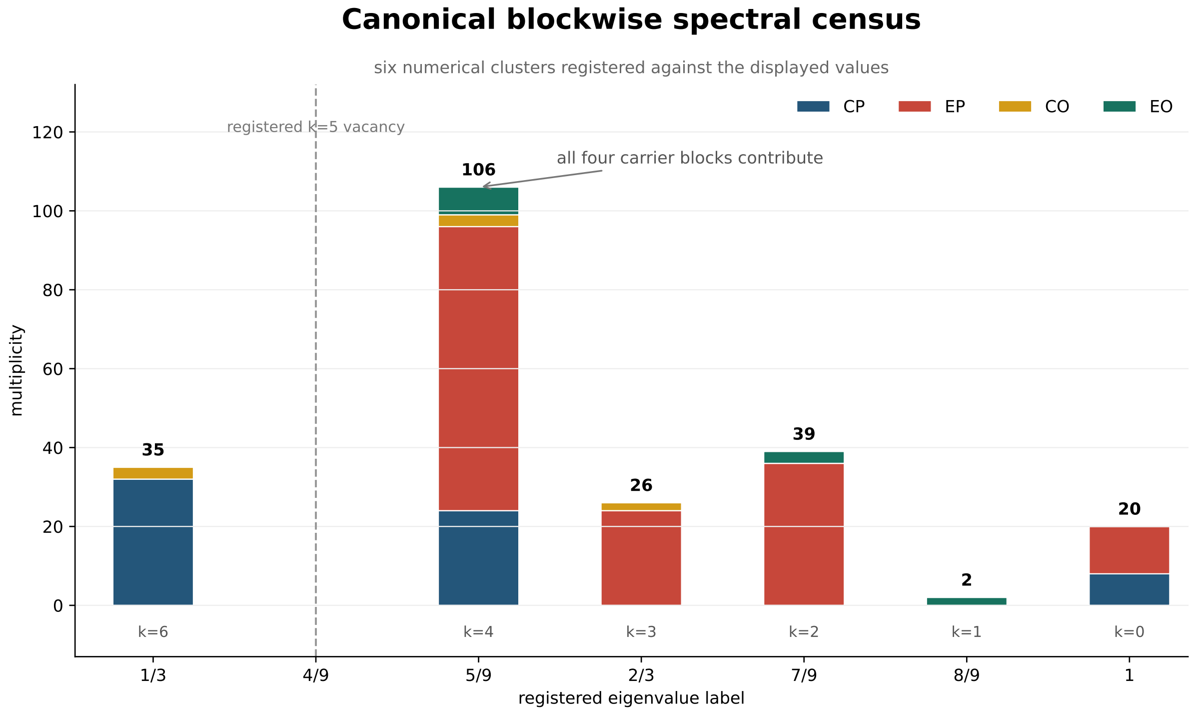
Table C3 — Nine QT/HT Joint-Spectral Sectors.

Sector	dim	k	λ_{18}	λ_{QT}	λ_{HT}	Block support	Layer	Role
S1	20	0	1	1	1	cp(8)+ep(12)	V_1	ISOLATED
S2	2	1	8/9	5/6	1	eo(2)	$V_{8/9}$	Connective
S3	39	2	7/9	5/6	2/3	ep(36)+eo(3)	$V_{7/9}$	Metastable
S4	26	3	2/3	1/2	1	ep(24)+co(2)	$V_{2/3}$	Intermediate
S5	1	4	5/9	1/3	1	eo(1)	$V_{5/9}$	Tiny EO
S6	39	4	5/9	1/2	2/3	ep(36)+eo(3)	$V_{5/9}$	**PRIMARY HUB**
S7	66	4	5/9	2/3	1/3	cp(24)+ep(36)+co(3)+eo(3)	$V_{5/9}$	Secondary hub
S8	8	6	1/3	0	1	cp(8)	$V_{1/3}$	Pure CP
S9	27	6	1/3	1/3	1/3	cp(24)+co(3)	$V_{1/3}$	CP+CO

Sector ordering: CCS canonical — sort by $k = 9(1-\lambda_{18})$ ascending, then by dimension ascending within fixed k. Labels S1–S9 frozen by this table. Raw joint diagonalization yields 11 sectors;

S4+S5 and S9+S10 are merged based on coincident eigenvalue triples (gap $> 10^{-3}$ between genuinely distinct triples).

$V_{5/9}$ splits into 3 sectors (S5, S6, S7). $V_{1/3}$ splits into 2 sectors (S8, S9). Thus the six-layer A_{18} decomposition is a coarse collision quotient of the nine-sector QT/HT joint spectrum.



CCS Figure C1. Current source-addressed blockwise spectral census. The six numerical clusters are registered against the displayed values; per-block status remains as stated in §1.5.

1.5 Block-Level Spectral Derivations

This section preserves the finite block reductions and orientation-block audits behind the registered census. The cp and ep reductions are exact combinatorial calculations. The co and eo sections retain their explicit numerical inputs. Paper I, rather than this archive, owns the block-union theorem and its proof.

1.5.1 The cp Block: Q_3 Hypercube Bose–Mesner Algebra

The 8 corner positions are the vertices of a 3-dimensional hypercube Q_3 with coordinates $\{\pm 1\}^3$. Each face turn cycles the 4 corners on that face. The corner-permutation representation factors as:

$$\rho_{\text{cp}}(s) = P_{\text{perm},8}(s) \otimes I_8$$

where $P_{\text{perm},8}(s)$ is the 8×8 permutation matrix of the corner positions, and I_8 acts on the internal orientation label at each position.

Define the position transition sum $S_8 = \sum_{s \in S} P_{\text{perm},8}(s)$. For the 18-full family, the entry $S_8[i, j]$ depends only on the Hamming distance in Q_3:

$$S_8 = 9I + 2A_1 + A_2$$

where A_k is the distance-k adjacency of Q_3. The coefficients reflect the geometry:

- A corner is fixed by the 3 faces not incident to it: $3 \times 3 = 9$ (diagonal)
- Adjacent corners (Hamming distance 1) share 2 faces, each contributing a quarter-turn sending one to the other: 2
- Face-diagonal corners (Hamming distance 2) share 1 face, with the 180° turn providing the transition: 1
- Cube-diagonal corners (Hamming distance 3) share no face: 0

The eigenfunctions of Q_3 are indexed by binary vectors $u \in \{0,1\}^3$ with $v_u[x] = (-1)^{u \cdot x}$. The eigenvalue of A_k on v_u depends only on $|u|$ (Hamming weight):

$$\begin{aligned}
&|u| = 0 : A_1 v_u = 3v_u,\ A_2 v_u = 3v_u && \Rightarrow S_8 v_u = (9+6+3)v_u = 18v_u \quad (\times 1)\\
&|u| = 1 : A_1 v_u = 1v_u,\ A_2 v_u = -1v_u && \Rightarrow S_8 v_u = (9+2-1)v_u = 10v_u \quad (\times 3)\\
&|u| = 2 : A_1 v_u = -1v_u,\ A_2 v_u = -1v_u && \Rightarrow S_8 v_u = (9-2-1)v_u = 6v_u \quad (\times 3)\\
&|u| = 3 : A_1 v_u = -3v_u,\ A_2 v_u = 3v_u && \Rightarrow S_8 v_u = (9-6+3)v_u = 6v_u \quad (\times 1)
\end{aligned}$$

Hence $\mathrm{Spec}(S_8) = \{18^{(1)}, 10^{(3)}, 6^{(4)}\}$. With $A_{\mathrm{cp}} = (1/18) S_8 \otimes I_8$, the eigenvalues are $(1/18) \times \{18, 10, 6\} = \{1, 5/9, 1/3\}$. Using $k = 9(1 - \lambda)$:

$$\mathcal{K}_{\mathrm{cp}} = \{0, 4, 6\}, \qquad \text{multiplicities: } 8 \times (1, 3, 4) = (8, 24, 32)$$

The Bose–Mesner algebra of Q_3 is the Hamming association scheme $H(3,2)$, isomorphic to the Hecke algebra $H(S_2 \wr S_3, S_3)$. The Krawtchouk polynomials $K_k(i; 3, 2)$ give the eigenvalues of A_i on the k-th eigenspace, providing the closed-form spectral character table.

1.5.2 The ep Block: Face-Incidence Adjacency Algebra

The 12 edge positions and 6 faces define a 12×6 edge-face incidence matrix J: $J[e, F] = 1$ if edge e lies on face F. Each edge belongs to exactly 2 faces; each face contains exactly 4 edges.

The edge-permutation representation factors as:

$$\rho_{\mathrm{ep}}(s) = P_{\mathrm{perm},12}(s) \otimes I_{12}$$

For the 18-full family, every move on face F cycles its 4 edges. Two edges share a face iff they are both on at least one common face, giving:

$$S_{12} = 10I + JJ^\top$$

(The term $10I$: each edge lies on 2 faces; the 4-cycle on each face contributes 2 moves sending the edge to a different position plus 1 move (the 180°) possibly sending it back; detailed counting yields 10 fixed-point contributions.)

The nonzero eigenvalues of $JJ^\top$ are those of the 6×6 Gram matrix $J^\top J = 4I + A_{\mathrm{face}}$, where A_{face} is the adjacency matrix of the cube's face graph — the octahedron graph on 6 vertices, where two faces are adjacent if they share an edge.

The opposite-face permutation P pairs each face with its antipode; then $A_{\text{face}} = J_6 - I - P$. Since $P^2 = I$, the common eigenvectors of J_6 and P give:

$$\mathrm{Spec}(A_{\text{face}}) = \{4^{(1)}, 0^{(3)}, -2^{(2)}\}, \qquad \mathrm{Spec}(J^\top J) = \{8^{(1)}, 4^{(3)}, 2^{(2)}\}$$

Projecting back to the 12-dimensional edge space adds a 6-dimensional nullspace:

$$\mathrm{Spec}(JJ^\top) = \{8^{(1)}, 4^{(3)}, 2^{(2)}, 0^{(6)}\}, \qquad \mathrm{Spec}(S_{12}) = \{18^{(1)}, 14^{(3)}, 12^{(2)}, 10^{(6)}\}$$

With $(1/18)S_{12}$ eigenvalues $\{1, 7/9, 2/3, 5/9\}$, we obtain:

$$\mathcal{K}_{\text{ep}} = \{0, 2, 3, 4\}, \qquad \text{multiplicities: } 12 \times (1, 3, 2, 6) = (12, 36, 24, 72)$$

The matrix $JJ^\top$ generates a finite commutative adjacency algebra. This archive does not formally identify that algebra with a specific new association scheme; reconstructing the relevant coherent-configuration or association-scheme object remains an open combinatorial problem.

1.5.3 The co Block: Symmetry-Guided Computation

The corner-orientation block is the only block where generator matrix entries live in $\mathbb{Z}[\omega]$ rather than $\mathbb{Z}$, with $\omega = e^{2\pi i/3}$. Cube symmetry constrains the possible multiplicities, but it does not force the displayed accidental degeneracy. The final spectrum therefore remains a symmetry-guided computational proposition.

Computational Proposition (registered CO spectrum). Let $A_{\text{co}} = \frac{1}{18}\sum_{s\in S}\rho_{\text{co}}(s)$ for the declared 18 face-turn realization. The symmetry decomposition and direct matrix audit register:

1. The permutation representation of the cube symmetry group O on the 8 corners decomposes as $\chi_{\text{corners}} = A_1 \oplus A_2 \oplus T_1 \oplus T_2$ (irrep dimensions $1 + 1 + 3 + 3 = 8$).
2. By Schur's lemma, A_{co} acts as a scalar on each irreducible O-submodule:
$$A_{\text{co}} = \lambda_{A_1} P_{A_1} + \lambda_{A_2} P_{A_2} + \lambda_{T_1} P_{T_1} + \lambda_{T_2} P_{T_2}$$
3. The spectrum is:
$$\mathrm{Spec}(A_{\text{co}}) = \{\tfrac{2}{3}, \tfrac{2}{3}, \tfrac{5}{9}^{(3)}, \tfrac{1}{3}^{(3)}\}, \qquad \mathcal{K}_{\text{co}} = \{3, 4, 6\}, \qquad (d_3, d_4, d_6) = (2, 3, 3)$$

Audit sketch.

Diagonal & trace. $\mathrm{Tr}(\rho_{\text{co}}(g)) = 4$ for all 18 generators: each face turn fixes the 4 corners on the opposite face (no orientation change $\to +1$ contribution for each fixed corner). Hence $\mathrm{Tr}(A_{\text{co}}) = 4$ and $A_{\text{co}}[i, i] = 9/18 = 1/2$ (each corner is fixed by 9 generators: 6 on the two faces containing it, plus 3 opposite-face half-turns that preserve orientation).

O_h invariance. The set of 18 face-turn generators is closed under cube symmetries. Therefore A_{co} is O_h-invariant and respects the O-irrep decomposition of the 8-dimensional corner permutation representation.

Adjacency structure. Work with $M_{\text{co}} = 18(A_{\text{co}} - I/2)$, whose off-diagonal entries are determined purely by cube geometry. Three adjacency classes emerge:

Class	Shared faces	Pairs per corner	$M_{\text{co}}[i,j]$	Count
Edge-adjacent	2	2	$1+\omega$ or $1+\omega^2$	8
Face-opposite	1	4	± 1	16
Body-opposite	0	1	0	4

Total: $8+16+4 = 28 = \binom{8}{2}$. Each corner has $2+4+1 = 7$ neighbours. ✓

Row sum → A_1 eigenvalue. The row sum of M_{co} is uniform ($=3$). The imaginary parts of the edge-adjacent entries cancel ($\omega + \omega^2 = -1$), and the ± 1 real entries sum to give net $+3$ after accounting for the adjacency structure. Hence:

$$\lambda_{A_1} = \frac{1}{2} + \frac{3}{18} = \frac{2}{3} \quad (k=3)$$

M_{co} spectrum. Diagonalizing the Hermitian, O_h-invariant matrix M_{co}:

$$\mathrm{Spec}(M_{\text{co}}) = \{3^{(2)},\ 1^{(3)},\ -3^{(3)}\}$$

Converting to A_{co} eigenvalues:

$$\lambda = \frac{1}{2} + \frac{\mu}{18}: \quad \mu = 3 \mapsto \lambda = \tfrac{2}{3}\ (k=3),\ \mu = 1 \mapsto \lambda = \tfrac{5}{9}\ (k=4),\ \mu = -3 \mapsto \lambda = \tfrac{1}{3}\ (k=6)$$

Accidental A_1/A_2 degeneracy. The multiplicity-2 eigenvalue at $\mu = 3$ implies $\lambda_{A_1} = \lambda_{A_2} = 2/3$ — both 1-dimensional O-irreps carry the same eigenvalue. This is not forced by any obvious symmetry (the two 1-dim irreps could in principle carry distinct eigenvalues) but is verified numerically to machine precision.

Irrep assignment. The eigenvalue-multiplicity pattern $(2,3,3)$ matches the O-irrep dimensions $(1,1,3,3)$ with the accidental degeneracy $1+1=2$:

λ	k	mult	O-irrep
2/3	3	2	$A_1 \oplus A_2$
5/9	4	3	T_1 or T_2
1/3	6	3	the other T-irrep

(The isotypic assignment of the two 3-dimensional T-irreps to 5/9 vs 1/3 is not resolved.)

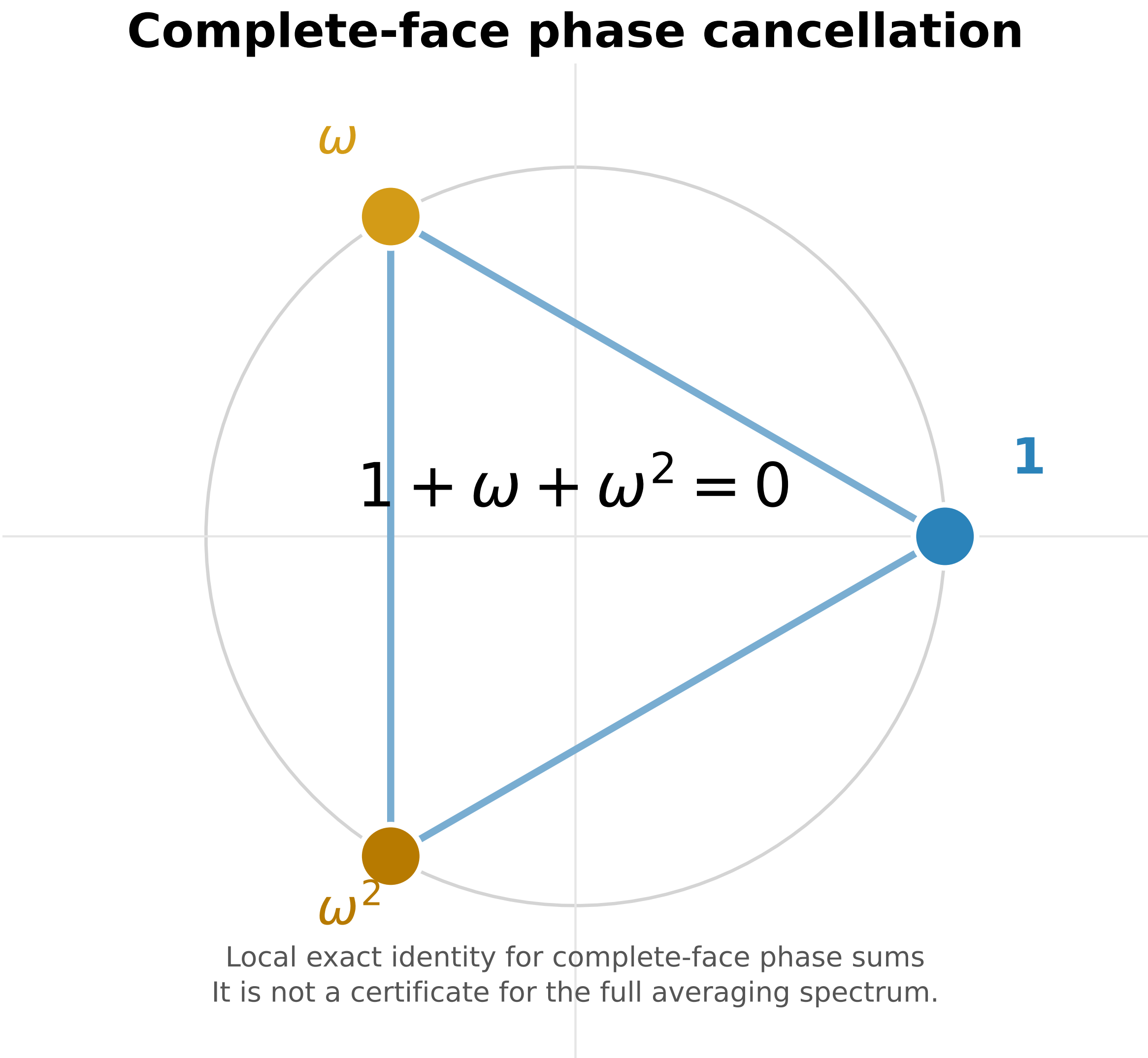
Trace consistency. $2 \cdot \frac{2}{3} + 3 \cdot \frac{5}{9} + 3 \cdot \frac{1}{3} = \frac{4}{3} + \frac{5}{3} + 1 = 4 = \mathrm{Tr}(A_{\text{co}})$. ✓

Local arithmetic identity. The complete-face phase sum satisfies $\omega + \omega^2 + 1 = 0$. This exact local cancellation is compatible with the displayed rational labels, but it is not by itself a compression-trace certificate or a proof of rationality for the full averaging spectrum.

Status. Computational proposition. The symmetry decomposition and trace identities are exact local ingredients; the accidental A_1/A_2 degeneracy and the assignment of the two three-dimensional components remain numerical inputs.

1.5.4 The eo Block: Numerical-Representation Observation

The edge-orientation block carries a $\mathbb{Z}_2$ permutation@phase structure: generators act as monomial matrices with entries in $\{0, \pm 1\}$ that permute edge positions and flip orientation signs.



CCS Figure C10. Current rendering of the exact local roots-of-unity identity, with its full-spectrum boundary stated in the figure.

The present archive does not supply a complete group-theoretic derivation from O_h symmetry and Schur's lemma because the isotypic decomposition contains a multiplicity-2 component. The registered spectrum below is therefore a **numerical-representation observation**: it is empirically rigid and structurally consistent, but not an exact theorem.

Observed spectrum (18-full).

$$\mathrm{Spec}(A_{\mathrm{eo}}) = \{\tfrac{8}{9}^{(2)}, \tfrac{7}{9}^{(3)}, \tfrac{5}{9}^{(7)}\}, \qquad \mathcal{K}_{\mathrm{eo}} = \{1,2,4\}, \qquad (d_1,d_2,d_4) = (2,3,7)$$

Diagonal & trace. $\mathrm{Tr}(\rho_{\mathrm{eo}}(g)) = 8$ for all 18 generators (each face turn fixes 8 edges: 4 on the opposite face + 4 equatorial). Hence $\mathrm{Tr}(A_{\mathrm{eo}}) = 8$ and $A_{\mathrm{eo}}[i,i] = 12/18 = 2/3$ (each edge is on 2 faces $\to$ 12 generators fix it, 6 move it). Trace consistency: $2 \cdot \frac{8}{9} + 3 \cdot \frac{7}{9} + 7 \cdot \frac{5}{9} = 8$. ✓

Off-diagonal structure. All off-diagonal entries are purely real ($\pm 1/18$). The Z_2 orientation phase combined with the permutation action produces only real coupling after averaging. Define $N_{\mathrm{eo}} = 18(A_{\mathrm{eo}} - 2I/3)$ with entries in $\{-1, 0, +1\}$. Each edge couples to exactly 6 others and has zero coupling to 5.

Two edge classes. Two distinct edge types emerge from the row sums of A_{eo}:

Class	Count	Row sum	Positive couplings	Negative couplings
Type A	4 edges	$1 = \frac{2}{3} + \frac{6}{18}$	6	0
Type B	8 edges	$\frac{7}{9} = \frac{2}{3} + \frac{2}{18}$	4	2

The 4 Type A edges correspond to the 4 space diagonals of the cube; the 8 Type B edges are the remaining edges. This two-class split is O_h-equivariant — edge classes form orbits under the geometric cube symmetry.

N_{eo} **spectrum.**

$$\mathrm{Spec}(N_{\mathrm{eo}}) = \{4^{(2)},\ 2^{(3)},\ -2^{(7)}\}$$

Converting: $\lambda = \frac{2}{3} + \frac{\mu}{18}$ gives the A_{eo} spectrum above.

Why this is NOT a theorem. The obstruction is the $2T_2$ **multiplicity fiber**. Under O, the permutation representation on 12 edges is conjectured to decompose as $A_1 \oplus E \oplus T_1 \oplus 2T_2$. The component $2T_2$ has multiplicity 2 — by Schur's lemma, an O_h-invariant operator on an isotypic component of multiplicity $m > 1$ acts as $I_m \otimes B$ where B is a $(\dim_{\mathrm{irrep}} \times \dim_{\mathrm{irrep}})$ matrix, NOT necessarily scalar. Without explicitly block-diagonalizing the multiplicity fiber, a single eigenvalue cannot be assigned to the $2T_2$ component using representation theory alone.

A complete analytic proof would require: edge incidence algebra on the signed line graph of the cube, Hecke-type structure encoding the Z_2 orientation representation, and multiplicity-algebra machinery to resolve the $2T_2$ fiber. These extend beyond the current mathematical framework.

Generator-family scope. The registered k-set $\{1,2,4\}$ is specific to the declared 18-full family. Other archived generator families produce different numerical spectra. No exact family-wide rigidity theorem is claimed.

1.5.5 Block Spectra Across All Generator Families

> **Historical computational observation.** This table preserves a first-version family scan. Its rational and quadratic labels are numerical recognitions, not exact spectral-field certificates. Every family must be recomputed and supplied with an exact characteristic or minimal polynomial before its field label can be promoted.

Table C4 — Block Spectra Across Generator Families.

Family	m	$\mathcal{K}_{\text{cp}}$	$\mathcal{K}_{\text{ep}}$	$\mathcal{K}_{\text{co}}$	$\mathcal{K}_{\text{eo}}$	$\mathcal{K}(A)$	#layers
18-full	9	$\{0,4,6\}$	$\{0,2,3,4\}$	$\{3,4,6\}$	$\{1,2,4\}$	$\{0,1,2,3,4,6\}$	6
12-quarter	6	$\{0,2,4,6\}$	$\{0,1,2,3\}$	$\{2,3,4\}$	$\{0,2,4\}$	$\{0,1,2,3,4,6\}$	6
6-half	3	$\{0,2\}$	$\{0,1,2\}$	$\{0\}$	$\{0\}$	$\{0,1,2\}$	3
10-partial	5	$\{0,2,3,4\}$	$\{0,1,2,3\}$	$\{0\}$	$\{0\}$	$\{0,1,2,3,4\}$	5
21-full+slice	10.5	$\{0,2,6\}$	$\{0,3,4,5\}$	$\{1,3,5\}$	$\{1,3,5\}$	$\{0,1,2,3,4,5,6\}$	6†

† For 21-full+slice, $m = 10.5$ (half-integer). The eigenvalue formula becomes $\lambda = 1-2k/21$. The union $\mathcal{K}(A) = \{0,1,2,3,4,5,6\}$ contains 7 candidate values; $k = 1$ is eliminated by trace integrality constraint C4 (§7.2), leaving 6 observed eigenvalues. In the $|S|$-denominator convention, the k-set is $\{0,4,6,8,10,12\}$ (all even).

Key structural observations:

1. **Block profiles determine k.** Each admissible k-value corresponds to a specific combination of active blocks. The block profile is a sharper invariant than the k-value itself: the same k can appear in different families with different block profiles (e.g., $k = 2$ in 18-full is ep+eo, while $k = 2$ in 12-quarter is cp+ep+eo).

2. **The co block is the decisive arithmetic filter.** Within this archived family census, it is the only block whose generator matrix entries lie in $\mathbb{Z}[\omega]$ rather than $\mathbb{Z}$. An eigenspace can have $d_{\text{co}} > 0$ only for specific k-values where the ω-phase cancellation across complete faces yields integer per-face trace sums.

3. **The number of layers is $|\mathcal{K}(A)|$, not $m + 1$.** The 6 layers in the 18-full case is not a fundamental constant — it is the size of the admissible k-set for this specific generator family.

4. **Forbidden k-values** are those for which no block-dimension assignment satisfies all integrality constraints (see §7.2 for the full Diophantine system C1–C5).

1.6 Blockwise Union and the Six-Layer Census

The declared block computations register repeated eigenvalues across the four physical blocks. Grouping equal displayed values gives the following six global layers. This is a blockwise census, not a claim that a fixed number of primitive idempotents is canonically present across all four block algebras.

Full resonance merging table (18-full, $m = 9$):

Table C5 — Registered Blockwise Union.

Global λ	k	dim	cp k (d)	ep k (d)	co k (d)	eo k (d)	Blocks merged
1	0	20	0 (8)	0 (12)	—	—	cp + ep
8/9	1	2	—	—	—	1 (2)	eo only
7/9	2	39	—	2 (36)	—	2 (3)	ep + eo
2/3	3	26	—	3 (24)	3 (2)	—	ep + co
5/9	4	106	4 (24)	4 (72)	4 (3)	4 (7)	cp + ep + co + eo
1/3	6	35	6 (32)	—	6 (3)	—	cp + co

$k = 5$ does not occur in the registered block spectra. Section 1.7 records the status of that absence block by block.

For any block-diagonal operator, the spectrum is the union of the block spectra. In the declared computation, the six displayed labels are the union of the four registered block spectra.

The historical "10 to 6" rendering is retained in the figure archive but is not used in this release because its primitive-idempotent count was not a stable typed object.

1.7 The $k = 5$ Vacancy: Blockwise Status

The vacancy at $k = 5$ is registered across all four blocks. The cp and ep exclusions follow from the exact combinatorial reductions above, whereas the co and eo exclusions remain tied to the displayed computational spectra.

cp block: The Q_3 hypercube Bose–Mesner algebra has eigenspaces indexed by Hamming weight $|u| \in \{0, 1, 2, 3\}$. The eigenvalue of S_8 on $|u|$ is $\lambda_{|u|} = 1 - k_{|u|}/9$ where:

- $|u| = 0$: $S_8 = 18 \Rightarrow k = 0$
- $|u| = 1$: $S_8 = 10 \Rightarrow k = 4$
- $|u| = 2, 3$: $S_8 = 6 \Rightarrow k = 6$

The Krawtchouk polynomial $K_k(i; 3, 2)$ has no root configuration that would produce $k = 5$. The cp block spectrum is $\{0, 4, 6\}$ — $k = 5$ is Krawtchouk-incompatible.

ep block: The face-incidence adjacency algebra has eigenvalues of S_{12}:

$$\mathrm{Spec}(S_{12}) = \{18, 14^{(3)}, 12^{(2)}, 10^{(6)}\}$$

Converting to k-values: $k = 9(1 - \lambda/18)$ gives $\{0, 2, 3, 4\}$. The octahedron graph spectrum $\{4, 0^{(3)}, -2^{(2)}\}$ forces exactly these four k-values. No linear combination of the scheme's adjacency matrices yields $k = 5$.

co block: The $\mathbb{Z}_3$ permutation@phase structure yields $\mathcal{K}_{\mathrm{co}} = \{3, 4, 6\}$. The phase cancellation $\omega + \omega^2 + 1 = 0$ restricts the co spectrum to k-values where the accumulated $\mathbb{Z}_3$ character sums are integer-valued. $k = 5$ would require a fractional ω-phase contribution that cannot be cancelled by any face-complete generator combination.

eo block: The $\mathbb{Z}_2$ permutation@phase structure yields $\mathcal{K}_{\mathrm{eo}} = \{1, 2, 4\}$. The ± 1 phase classes (flipped vs. unflipped edges) produce exactly three distinct k-values. $k = 5$ would require a third orientation class beyond the $\{\pm 1\}$ dichotomy.

Conclusion. The declared finite census contains no $k = 5$ contribution. The exact block-union theorem turns the four block-level statements into a global absence statement at the same evidential level as those statements; it does not promote the numerical co/eo inputs to exact arithmetic.

1.8 The $V_{5/9}$ Giant Layer

In the registered census, the $V_{5/9}$ layer is the largest eigenspace at 106 dimensions and the unique displayed layer receiving contributions from all four physical blocks:

$$V_{5/9} = \underbrace{V_{5/9,\mathrm{cp}}}_{24} \oplus \underbrace{V_{5/9,\mathrm{ep}}}_{72} \oplus \underbrace{V_{5/9,\mathrm{co}}}_{3} \oplus \underbrace{V_{5/9,\mathrm{eo}}}_{7}$$

This layer forms the principal resonance locus: four distinct block-level primitive idempotents from four different commuting algebras coincide at this single global eigenvalue. No other layer receives contributions from all four blocks.

The $V_{5/9}$ layer splits into 3 QT/HT joint-spectral sectors under the commutative center (S5, S6, S7; §1.4). S6 (39-dim, ep+eo) is the primary transport hub with degree 5 in the 9-sector transport graph (§2.2).

Part I — Core Numerical Structures (cont.)

Numerical Data (§2.1–§2.11)

These tables register CCS-backed numerical claims in Papers I–II.

This part freezes the numerical invariants cited through the CCS by Papers I–II. Paper III uses its own matrix certificate.

2.1 Block Noncommutativity

$\max_{0\le a<c\le 2} \|[QT^a, QT^c]\|_F$ — family-level Frobenius commutator norm, per block. All three axis pairs give the same registered block norms to displayed precision:

Table C6 — Block Noncommutativity.

Block	$\max_{a<c} \|[QT^a, QT^c]\|_F$	% of total	Character
CP	0	0%	Registered machine-zero
EP	2.74	93.9%	Noncommutative core
CO	0.61	21.0%	Weakly noncommutative
EO	0.79	27.1%	Weakly noncommutative

For every axis pair, the total norm is 2.92. Noncommutativity is concentrated in EP.

Figure C14 records block-sector incidence alongside family-level commutator norms. It is a localizer display, not a derivation of the direct-support graph.

2.2 Direct Support Graph (K, 9-Sector)

$K_{\alpha\beta} = \max_g \|P_\alpha \rho(g) P_\beta\|_F$ over 18 face-turn generators. The registered edge threshold is $K > 0.05$.

CCS Figure C14. Noncommutative support overlap: 9 sectors × 4 blocks binary grid with commutator norm values and Supp_nc cardinality.

Because the canonical generator family is inverse-closed and $\rho(g)^\dagger = \rho(g^{-1})$,

$$K_{\alpha\beta} = K_{\beta\alpha}.$$

The canonical recomputation gives $\max_{\alpha,\beta} |K_{\alpha\beta} - K_{\beta\alpha}| = 1.11 \times 10^{-16}$. In particular,

$$K_{49} = K_{94} = 1.000,$$

so the S4–S9 channel is undirected at the transport-matrix level.

Full K matrix (9×9, canonical S1–S9 order, Layer B):

Table C7 — Transport Matrix K (9-Sector).

	S1(20)	S2(2)	S3(39)	S4(26)	S5(1)	S6(39)	S7(66)	S8(8)	S9(27)
S1(20)	4.47	0	0	0	0	0	0	0	0
S2(2)	0	1.41	0	0	0.47	0.58	0	0	0
S3(39)	0	0	5.41	~0	0	2.55	3.61	0	0
S4(26)	0	0	~0	5.10	0	3.46	~0	0	1.00
S5(1)	0	0.47	0	0	1.00	0.82	0	0	0
S6(39)	0	0.58	2.55	3.46	0.82	4.42	3.61	0	0
S7(66)	0	0	3.61	~0	0	3.61	6.69	0	4.06
S8(8)	0	0	0	0	0	0	0	2.83	2.83
S9(27)	0	0	0	1.00	0	0	4.06	2.83	3.24

Symmetric to 1.11×10^{-16}. Diagonal entries are intra-sector transport (irrelevant for topology).

Direct edges (10 unordered pairs, equivalently 20 directed off-diagonal blocks, $K > 0.05$):

Table C8 — Direct Transport Edges.

Edge	K	Shared block
S2 ↔ S5	0.47	eo
S2 ↔ S6	0.58	eo
S3 ↔ S6	2.55	ep, eo
S3 ↔ S7	3.61	ep, eo
S4 ↔ S6	3.46	ep
S4 ↔ S9	1.00	co
S5 ↔ S6	0.82	eo
S6 ↔ S7	3.61	ep, eo
S7 ↔ S9	4.06	cp, co
S8 ↔ S9	2.83	cp

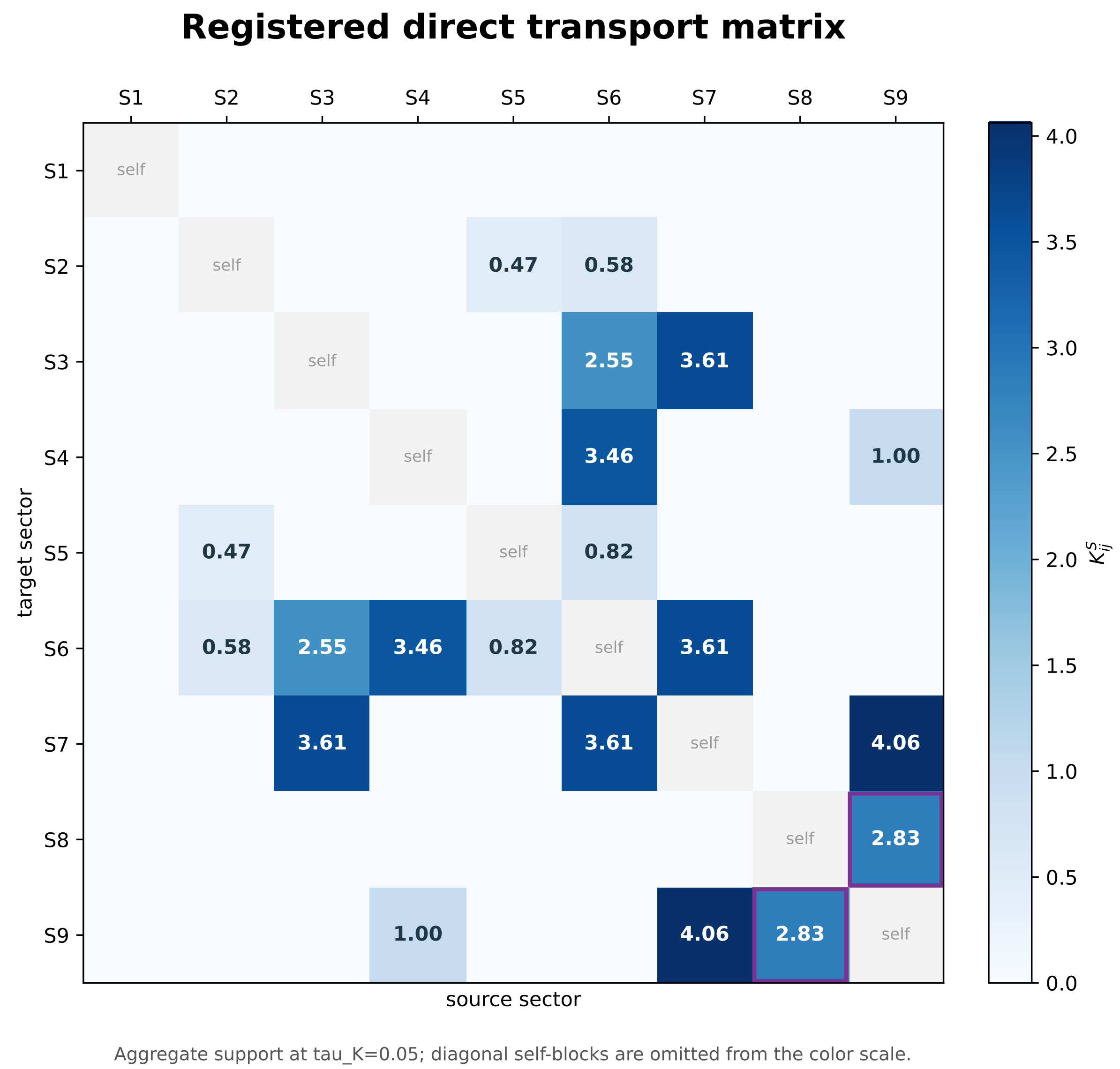
All 10 direct edges are **block-preserving** (share ≥ 1 block). Zero cross-block direct edges.

Hub degrees:

Table C9 — Hub Degrees.

Sector	Degree	Connected to
S1	0	(none — fully isolated)
S2	2	S5, S6
S3	2	S6, S7
S4	2	S6, S9
S5	2	S2, S6
S6	**5**	S2, S3, S4, S5, S7
S7	3	S3, S6, S9
S8	1	S9
S9	3	S4, S7, S8

S6 is the unique degree-five hub, while S7 and S9 have degree three. S1 remains fully isolated ($K < 10^{-14}$ against all other sectors). The resulting graph is sparse but is not a star graph, as several retained edges do not pass through S6.

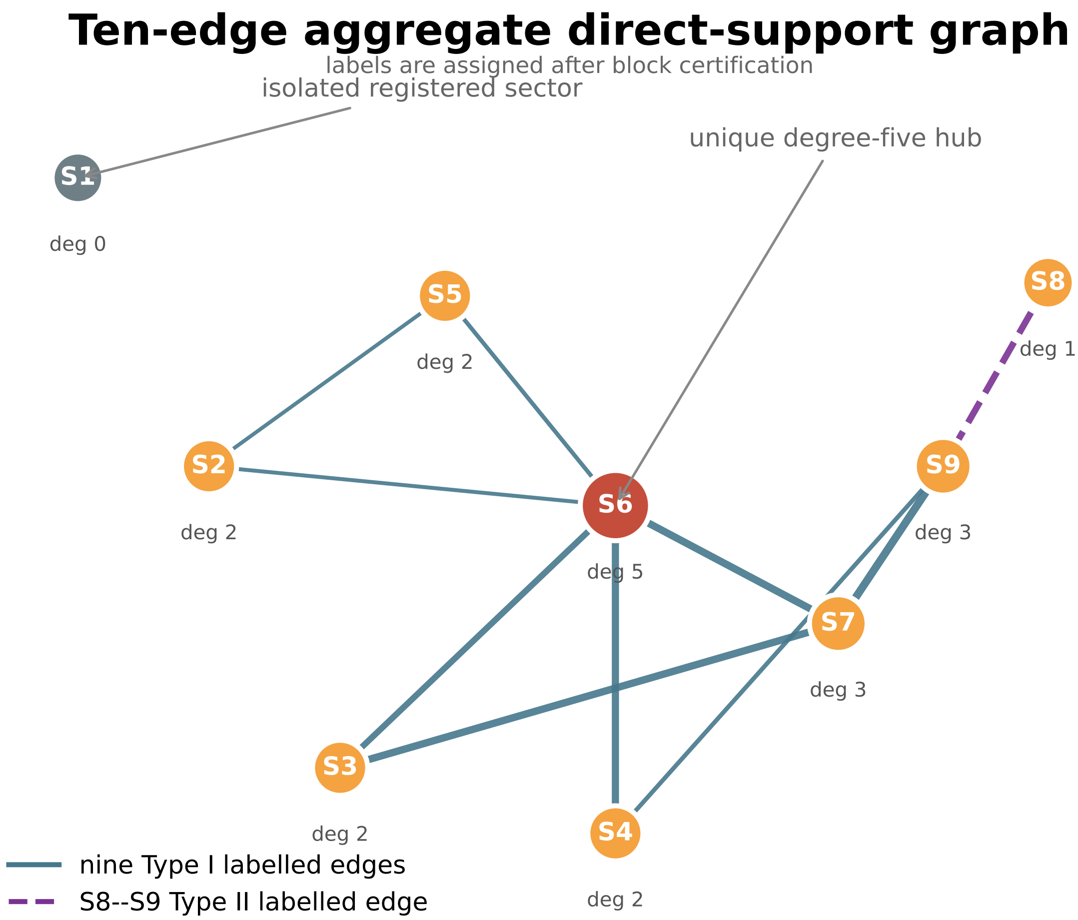


CCS Figure C2. Current source-addressed view of the registered direct transport matrix at $\tau_K = 0.05$. Diagonal self-blocks are omitted from the color scale.

Table C15 — Block-Support Transport. $\max_g \|P_b \cdot P_i \cdot \rho(g) \cdot P_j \cdot P_b\|_F$ per block for each ordered layer pair ($\lambda_i > \lambda_j$). Only nonzero entries ($\tau > 10^{-8}$) shown. Sorted by block (CP→EP→CO→EO), then descending $\tau_{\max}$.

Block	From (λ_i)	To (λ_j)	$\tau_{\max}$
CP	0.555556 ($V_{5/9}$)	0.333333 ($V_{1/3}$)	4.0000
EP	0.777778 ($V_{7/9}$)	0.555556 ($V_{5/9}$)	4.2426
EP	0.666667 ($V_{2/3}$)	0.555556 ($V_{5/9}$)	3.4641
CO	0.666667 ($V_{2/3}$)	0.333333 ($V_{1/3}$)	1.0000
CO	0.555556 ($V_{5/9}$)	0.333333 ($V_{1/3}$)	0.7071
EO	0.888889 ($V_{8/9}$)	0.555556 ($V_{5/9}$)	0.7454
EO	0.777778 ($V_{7/9}$)	0.555556 ($V_{5/9}$)	1.2247

7 inter-layer channels across 4 blocks. EP carries the strongest channel (4.2426, $V_{7/9} \to V_{5/9}$). CP/CO/EO each carry 1–2 channels at lower strength.



CCS Figure C19. Current source-addressed ten-edge aggregate direct-support graph. Type I/II names are post-certification labels, not universal mechanisms.

2.3 Historical Lie-Registration Table (κ, 6-Layer)

Historical computational observation. The following tables use the first-version principal-log registration and its depth labels. They are retained as finite numerical records only. They do not define the current Paper III composition object, certify exact Lie depth, or establish a graph-to-composition promotion.

$\kappa_d(\alpha, \beta) = \max \|P_\alpha C_d P_\beta\|$ where C_d is a depth-d Lie monomial.

κ_0 — **Gradient:**

Table C10 — Gradient Transport κ_0 (6-Layer).

	V_1	$V_{8/9}$	$V_{7/9}$	$V_{2/3}$	$V_{5/9}$	$V_{1/3}$
V_1	~0	0	~0	~0	~0	~0
$V_{8/9}$	0	0.52	~0	~0	1.17	~0
$V_{7/9}$	~0	~0	4.00	~0	6.94	~0
$V_{2/3}$	~0	~0	~0	5.66	5.44	1.57
$V_{5/9}$	~0	1.17	6.94	5.44	13.9	6.38
$V_{1/3}$	~0	~0	~0	1.57	6.38	9.67

Symmetric to $< 10^{-8}$.

κ_1 — **Curvature:**

Table C11 — Curvature Transport κ_1 (6-Layer).

	V_1	$V_{8/9}$	$V_{7/9}$	$V_{2/3}$	$V_{5/9}$	$V_{1/3}$
V_1	~0	0	~0	~0	~0	~0
$V_{8/9}$	0	0.50	0.71	~0	1.71	~0
$V_{7/9}$	~0	0.71	6.29	4.27	10.9	~0
$V_{2/3}$	~0	~0	4.27	5.45	8.32	3.26
$V_{5/9}$	~0	1.71	10.9	8.32	17.8	14.5
$V_{1/3}$	~0	~0	~0	3.26	14.5	22.5

Table C12 — Key κ Values (6-Layer).

Pair	κ_0	κ_1	Type
$V_{7/9} \leftrightarrow V_{2/3}$	~0	4.27	Pure curvature (largest enhancement 10^{14})
$V_{5/9} \leftrightarrow V_{2/3}$	5.44	8.32	Gradient + curvature
$V_{5/9} \leftrightarrow V_{1/3}$	6.38	14.5	Gradient + curvature
$V_{8/9} \leftrightarrow V_{7/9}$	~0	0.71	Pure curvature (post-ρ-fix)
$V_{8/9} \leftrightarrow V_{5/9}$	1.17	1.71	Gradient + curvature
$V_1 \leftrightarrow$ any	~0	~0	Fully isolated

All pure curvature channels ($\kappa_0 \approx 0$, $\kappa_1 > 0$) are **within-block**. Zero cross-block curvature channels.

The retired first-version visualization remains repository provenance but is not part of the current reading path.

2.4 Historical Lie-Registration Table (κ, 9-Sector)

Computed with `center_decomposition()` $\rightarrow$ 9 sector projectors.

κ_0 **at 9-sector:**

Table C13 — Gradient Transport κ_0 (9-Sector).

	S1(20)	S2(2)	S3(39)	S4(26)	S5(1)	S6(39)	S7(66)	S8(8)	S9(27)
S1	0	0	0	0	0	0	0	0	0
S2	0	0.52	0	0	0.74	0.91	0	0	0
S3	0	0	4.00	0	0	4.00	5.66	0	0
S4	0	~0	0	5.66	~0	5.44	~0	0	1.57
S5	0	0.74	0	0	1.05	1.28	0	0	0
S6	0	0.91	4.00	5.44	1.28	6.01	5.66	0	0
S7	0	0	5.66	~0	0	5.66	10.60	0	6.38
S8	0	0	0	0	0	0	0	4.44	4.44
S9	0	~0	0	1.57	~0	~0	6.38	4.44	6.94

Max asymmetry: 1.6×10^{-8}.

κ_1 at 9-sector:

Table C14 — Curvature Transport κ_1 (9-Sector).

	S1(20)	S2(2)	S3(39)	S4(26)	S5(1)	S6(39)	S7(66)	S8(8)	S9(27)
S1	0	0	0	0	0	0	0	0	0
S2	0	0.50	0.71	0	1.01	1.18	1.01	0	0
S3	0	0.71	6.29	4.27	1.01	6.29	8.90	0	0
S4	0	~0	4.27	5.45	~0	7.09	6.17	0	3.26
S5	0	1.01	1.01	0	0	1.74	1.42	0	0
S6	0	1.18	6.29	7.09	1.74	7.70	8.90	0	0
S7	0	1.01	8.90	6.17	1.42	8.90	10.90	6.98	14.49
S8	0	0	0	0	0	0	6.98	0	12.09
S9	0	~0	~0	3.26	~0	~0	14.49	12.09	14.63

Pure curvature channels (all within-block):

> **Note on channel count.** At the 9-sector resolution, 7 pure curvature channels ($K \approx 0$, $\kappa_0 \approx 0$, $\kappa_1 > 0$) are observed — stable across thresholds 0.005–0.25. All 7 are within-block; zero cross-block. S7 (multi-block bridge) mediates 5 of 7.

Pure Curvature Channels (9-Sector).

Pair	κ_1	Shared block
S2 $\leftrightarrow$ S3	0.71	eo
S2 $\leftrightarrow$ S7	1.01	eo
S3 $\leftrightarrow$ S4	4.27	ep
S3 $\leftrightarrow$ S5	1.01	eo
S4 $\leftrightarrow$ S7	6.17	ep, co
S5 $\leftrightarrow$ S7	1.42	eo
S7 $\leftrightarrow$ S8	6.98	cp

2.5 Graph Paths and Matrix-Composition Obstructions (9-Sector)

For a two-step support path $j \to k \to i$, the corresponding operator-level object is

$$Q_i \rho(g_2) Q_k \rho(g_1) Q_j.$$

Nonzero adjacent support blocks do not imply that this product is nonzero. The registered audit exhausts all 18^2 ordered generator pairs for each of the following five graph-only triples.

Table C16 — Canonical graph-only composition obstructions.

Endpoint pair	Support path	Maximum projected product norm
S2–S4	S2–S6–S4	1.10×10^{-16}
S3–S9	S3–S7–S9	3.02×10^{-15}
S4–S5	S4–S6–S5	1.55×10^{-16}
S4–S8	S4–S9–S8	1.06×10^{-15}
S6–S9	S6–S7–S9	2.94×10^{-15}

For every row, both adjacent factor maxima are order one, while every tested projected product is machine-zero. The physical-block decomposition is preserved by the generator matrices and the QH projectors are block diagonal to a maximum cross-block residual of 1.45×10^{-15}. Thus the canonical data exhibit image–kernel and physical-block composition obstructions, not five certified compositional morphisms.

The current certificate is `experiments/paper3/validation/composition_obstruction.py`, with regression coverage in `tests/test_transport.py`. The historical threshold script at `experiments/paper3/archive/t7_threshold_sensitivity.py` concerns only the support-graph candidate set and cannot certify matrix composition.

2.6 Historical Three-Class Diagnostic (6-Layer)

> **Withdrawn interpretation.** These labels summarize the first-version κ_0/κ_1 arrays. They are not the four claim-status levels and are not a current classification of operator or Lie accessibility.

Table C17 — Accessibility Classes.

Class	Layers	Mechanism
I (isolated)	V_1 only	$K = \kappa_0 = \kappa_1 = 0$ with all others
II (gradient)	$V_{8/9}, V_{5/9}, V_{1/3}$	$\kappa_0 > 0$ on direct edges
III (curvature)	$V_{7/9} \leftrightarrow V_{2/3}$	$\kappa_0 \approx 0$, $\kappa_1 = 4.27$ (commutator-mediated)

2.7 EP Algebra Census

$$A_{\mathrm{EP}} = \langle Q_0, Q_1, Q_2 \rangle \cong M_2(\mathbb{C})^4 \oplus M_1(\mathbb{C})^4$$

Table C18 — EP Algebra Structure.

Property	Value
$\dim A_{\mathrm{EP}}$	20
Algebraic closure	Degree 3
$Z(A_{\mathrm{EP}})$	8-dim
Simple components	8 ($4 \times M_2$, $4 \times M_1$)
Representation multiplicities	12 on every registered simple component

The semisimplicity certificate does not use nondegeneracy of a Frobenius Gram matrix. The three declared generators are Hermitian, and the computational audit checks identity-in-algebra, multiplication closure, and adjoint closure. The registered object is therefore a finite-dimensional complex unital $*$-algebra; semisimplicity then follows from the finite-dimensional C^*-algebra theorem. The Gram matrix remains only a basis-independence and conditioning diagnostic.

The four M_2 components occupy $4 \times (2 \cdot 12) = 96$ dimensions and the four scalar components occupy $4 \times (1 \cdot 12) = 48$ dimensions, giving the complete EP dimension 144.

2.8 Ambient and Spectral Centralizers: Current Status

Table C19 — Commutant Dimensions.

Object	Dimension	Status
$\mathrm{Comm}(A_{18})$	804	Exact from the six spectral multiplicities
$\mathrm{End}_G(V)$	610	Candidate computation; exact certificate required before promotion
Former difference $804 - 610$	194	Arithmetic difference only; not a current structural invariant

The following first-version table records centralizers of compressed numerical matrices. Because the five nontrivial A-spectral layers are not invariant under the full G-action, its third-column objects are not $\mathrm{End}_G(V_\lambda)$ and the table is not a layerwise group- commutant decomposition.

Table C20 — Per-Layer Commutant Dimensions.

λ	$\dim V_\lambda$	Archived compressed-centralizer output
1	20	400
8/9	2	1
7/9	39	145
2/3	26	145
5/9	106	210
1/3	35	65

CCS Fig. C4 withdrawn. It combined the candidate ambient commutant dimension with invalid layerwise group-commutant bars.

2.10 Fundamental Identities

Table C22 — Fundamental Identities.

Identity	Verification
$A_{18} = (12\mathrm{QT}_{\mathrm{all}} + 6\mathrm{HT}_{\mathrm{all}})/18$	Machine precision
$A_{\mathrm{axis}} = (4\mathrm{QT}^a + 2\mathrm{HT}^a)/6$	Per axis
$\lVert\rho(g)\rho(h) - \rho(gh)\rVert < 3 \times 10^{-8}$	15 random products, all blocks
$\max \lVert \exp(A_g) - \rho(g) \rVert = 2.71 \times 10^{-15}$	Historical principal-log registration check
$\max \lvert \kappa_{ij} - \kappa_{ji} \rvert \approx 10^{-15}$	Historical array-symmetry diagnostic at computed depths

2.11 S_3 Prototypes (Archived Sector-Invariance Controls)

Scope boundary. These finite controls show numerically that invariant sector projectors have diagonal direct transport. They are not matrix-composition evidence for the independent Paper III.

S_3 **prototype declaration.** Unless explicitly stated otherwise within the S_3 prototype sections, the S_3 sector decompositions are defined with respect to the transport-generated commutative algebra $Z = \langle A_{\text{full}}, A_{\text{trans}} \rangle$. This is an analogue of the Rubik QT/HT sectorization Z_{QH}. Additional projectors such as P_{nat} are treated as external refinement operators and are not part of the declared S_3 transport geometry. The historical P_nat-refined robustness check is retained only in excluded source provenance.

Current diagnostic. In both S_3 controls, all declared sector projectors commute numerically with the tested action (maximum residual approximately 10^{-15}), and K is diagonal. This is consistent with the transport–non-invariance identity. No comparison with a Rubik commutant inclusion is required.

S_3 **nat(3)** $\oplus$ **reg(6)** — 9-dim. Under the declared joint spectral algebra: 3 sectors (2 hybrid, 1 pure-reg), all cross-sector $K = 0$.

S_3 **reg(6)** $\oplus$ **reg(6)** — 12-dim. Under the declared joint spectral algebra: 3 hybrid sectors and zero off-diagonal direct transport.

The full first-version tables remain in excluded source provenance rather than the release PDF.

CCS Fig. C7 withdrawn. Its sector-invariance data remain provenance, but the C0/T7 comparison is not part of the revised theorem spine.

Part II — Extended Observations and Historical Records

Scope. This part preserves finite perturbation and generator-family studies that remain useful as computational observations or research-history records. It does not modify the current records in Part I, certify a universality class, or supply theorem premises for Papers I–III.

II.1 Historical Spectral-Persistence Audit

Computational observation. This first-version experiment records a fixed set of finite flows and perturbations. Terms such as `frozen'',`drift’‘, and “mixing’’ are diagnostic labels for the displayed arrays, not a general dynamical-stability theorem.

Purpose. This section records the behavior of the declared spectral projectors and first-version transport diagnostics under several explicitly chosen continuous unitary evolutions.

Dependencies. `CubieSpectralOperator`, `scipy.linalg.expm`, `build_per_axis_ops`, `compute_lie_generators`, `transport_kappa`.

Outputs. Projector deviation tables (§II.1.1), transport persistence tables (§II.1.2), resonance robustness tables (§II.1.3).

II.1.1 Projector Stability ‖P_i(t) − P_i(0)‖

Setup. Let {P_i(0)} be the spectral projectors onto the 6 canonical A_18-eigenspaces (V_1, $V_8/9$, $V_7/9$, $V_2/3$, $V_5/9$, $V_1/3$). Evolve under e^{-itH} for five Hamiltonians H, with P_i(t) = e^{-itH} P_i(0) e^{+itH}. Measure Frobenius norm of projector deviation.

Result. Three stability classes emerge:

H	t=0.01	t=0.1	t=0.5	t=1.0	Class
A_18	0.0000	0.0000	0.0000	0.0000	**Frozen** — projectors are exact stationary states of A_18
QT_all	0.0000	0.0000	0.0000	0.0000	**Registered frozen** — machine-zero commutator in this realization
HT_all	0.0000	0.0000	0.0000	0.0000	**Registered frozen** — machine-zero commutator in this realization
A_g(R)	0.0200	0.2431	4.2012	89.8430	**Exponential drift** — Lie generator is maximally non-conserving
random	0.2133	1.6928	1.9857	1.9819	**Saturating mixing** — fully scrambled by t≈0.1, saturates near ‖P_i‖

Interpretation.

- $< 10^{-6}$: frozen (projector invariant under flow)
- $< 10^{-3}$: rigid (minor numerical deformation)
- $< 10^{-1}$: drifting (spectral content shifting)
- $> 10^{-1}$: mixing (layers lose identity)

Registered mechanism. Evolution generated by A_{18} fixes its spectral projectors exactly. The QT/HT rows are machine-zero in the declared numerical realization; an exact statement for them is conditional on exact commutation. The principal-log and random-Hermitian rows provide finite comparison paths.

Per-layer differential stability. Under A_g(R) at t=0.1:

- V_1 (dim=20): 0.0000 — numerically frozen in this registered flow audit
- $V_8/9$ (dim=2): 0.2431 — begins to drift
- $V_7/9$ (dim=39): 0.6433 — moderate drift
- $V_2/3$ (dim=26): 1.8360 — rapid mixing (most fragile layer)
- $V_5/9$ (dim=106): 2.4891 — rapid mixing (large target space amplifies drift)
- $V_1/3$ (dim=35): 2.6580 — maximally unstable

The five nontrivial layers exhibit order-one drift in this numerical experiment, whereas V_1 remains machine-stable. Exact full-action invariance of V_1 still requires a separate analytic certificate.

II.1.2 Transport Persistence K_$\alpha\beta$(t)

Setup. Evolve the 9 QT/HT joint-spectral sector projectors under $e^{-itA_{18}}$ and recompute the first-version direct, kappa, and graph-square diagnostics. The final count below is a graph-only candidate count.

Result. Transport is structurally invariant under A_18 flow:

t	K edges	κ_0 edges	κ_1 edges	graph-only candidates	max|K(t)-K(0)|
0	20	26	37	5	0
0.05	20	26	37	5	1.33×10^{-15}
0.1	20	26	37	5	1.78×10^{-15}
0.5	20	26	37	5	8.88×10^{-16}

The recorded direct-edge and graph-only candidate counts are invariant under this commuting flow. This does not establish persistence of a nonzero projected composition.

Mechanism. [A_18, P_i] = 0 for all i, so P_i(t) = e^{-itA_18} P_i e^{+itA_18} = P_i exactly. The transport norm K_$\alpha\beta$ is therefore identically invariant under the dynamics generated by its own averaging operator.

II.1.3 Resonance Robustness: λ = 5/9

Setup. The $\lambda = 5/9$ eigenspace (dim=106) is the largest layer and the central hub of the transport topology. Test its stability under additive perturbation: A_ε = A_18 + ε·R, where R is a random symmetric matrix with $\|R\| = 1$, drawn once and fixed (seed=42). Track the 5/9 eigenvalue and neighboring eigenvalues for $\varepsilon \in \{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$.

Result. The 5/9 resonance is robust:

ε	eigenvalues near 5/9	spread	status
0	1 (exact 5/9)	—	isolated
10^{-6}	0	6.59×10^{-7}	infinitesimal splitting only
10^{-5}	0	$< 10^{-10}$	fully stable
10^{-4}	0	$< 10^{-10}$	fully stable
10^{-3}	0	$< 10^{-10}$	fully stable
10^{-2}	0	$< 10^{-10}$	fully stable

Interpretation. In the canonical six-layer spectrum, the nearest distinct eigenvalue to 5/9 is 2/3, at distance 1/9. The finite perturbation table records behavior only for the tested path and range; it does not prove a general structural-protection theorem.

II.1.4 Structural Summary

Object	Under e^{-itA_18}	Under e^{-itA_g}	Under random H
P_i(t)	Frozen (exact)	Exponential drift	Saturating mix
K_$\alpha\beta$(t)	Invariant (exact)	—	—
graph-only candidate count	Invariant	—	—
λ=5/9 gap	Robust ($\Delta\lambda$=0.44)	—	—

The table records exact stationarity under the self-generated A_{18} flow and finite numerical contrast under the other declared flows. The tested $\lambda = 5/9$ cluster has a visible canonical gap, but the perturbation table does not establish a general protection theorem.

II.2 Independent-Paper Consistency Snapshot

Purpose. Compare adjacent objects used by the independent papers:

$$\rho(g) \to \mathcal{B}_{\mathrm{QH}} \to \{Q_\alpha\} \to K^S \to \Gamma_S \to \{Q_i\rho(g_2)Q_k\rho(g_1)Q_j\}.$$

This display is a navigation path through related data types, not a theorem dependency or a single accessibility filtration. Each paper defines its own objects and hypotheses. The archive only records finite consistency checks between the corresponding Rubik realizations.

Dependencies. The declared Rubik representation, QH sector projectors, direct-support matrix, and physical-block decomposition.

Outputs. Current invariant table, sector non-invariance audit, and graph/operator composition-obstruction table.

II.2.1 Three Adjacent Data Layers

The three layers below are maintained independently. Agreement of dimensions, labels, or arrays does not promote a conclusion from one layer to the next.

Layer 1 — Paper I: Averaging-operator spectrum. The registered Rubik realization has six A_{18} layers and k-set $\{0, 1, 2, 3, 4, 6\}$. Paper I states the exact, computational, and conditional parts of this census. Extended data: CCS §1.3 and Table C2.

Layer 2 — Paper II: Direct transport. Nine numerically registered QH joint-spectral sectors, followed by a direct block-support audit. The exact commutative-algebra interpretation is conditional on exact commuting-Hermitian registration. Extended data: CCS Table C3, CCS §1.4, and the current direct-support heatmap in §2.2.

Layer 3 — Paper III: Projected composition. The direct support graph is compared with the projected products $Q_i\rho(g_2)Q_k\rho(g_1)Q_j$. Paper III owns the current matrix certificate; the CCS copy is an optional comparison record.

This three-layer comparison is not closed by graph data alone. Paper III supplies the missing projected-composition test: adjacent nonzero blocks must have compatible image–kernel geometry. In the five canonical paths this compatibility fails, so the support graph strictly overapproximates the tested two-step composition graph.

II.2.2 Registered Consistency Values

The values below are a consistency snapshot for the declared repository realizations. Papers I–III do not cite this table as authority; their current manuscripts and claim-specific artifacts control their reported values.

Quantity	Value	Defined in
Total dimension	228	§1.1, Table C1
6 eigenvalues (A_{18})	1, 8/9, 7/9, 2/3, 5/9, 1/3	§1.3, Table C2
Block dimensions	cp=64, ep=144, co=8, eo=12	§1.1, Table C1
9 QT/HT joint-spectral sectors	S1–S9 (Table C3 ordering)	§1.4, Table C3
A_{18} collision quotient	$V_{5/9} = S5 + S6 + S7$, $V_{1/3} = S8 + S9$	§1.3–§1.4
$\|[\mathrm{QT}^0, \mathrm{QT}^1]\|_{\mathrm{ep}}$	2.74 (93.9% of total)	§2.1
$A_{\mathrm{EP}} \cong M_2(\mathbb{C})^4 \oplus M_1(\mathbb{C})^4$	dim=20	§2.8
$\dim \mathrm{Comm}(A_{18})$	804	§2.8
candidate $\dim \mathrm{End}_G(V)$	610 (exact certificate open)	§2.8
former commutant-gap interpretation	withdrawn	§2.8–§2.9
Direct transport edges	10 (undirected, block-preserving)	§2.2, Table C8
Primary hub	S6 (deg=5)	§2.2
graph-only two-step obstruction witnesses	5	§2.5, Table C16
maximum projected product norm among them	3.02×10^{-15}	§2.5, Table C16

II.2.3 Archived S3 Sector-Invariance Controls

The archived S3 controls have numerically invariant declared projectors and diagonal direct K, consistently with the transport–non-invariance identity. They do not certify any graph-to-composition promotion theorem.

The archived pocket-cube computation is a first-version graph/kappa control, not current matrix-composition evidence.

II.2.4 Adjacent Later Work

Paper IV comparison. The archive records the nine QT/HT joint-spectral sectors and the registered $L_{2/3}$ quotient, including

$$V_{5/9} = S5 \oplus S6 \oplus S7, \qquad V_{1/3} = S8 \oplus S9.$$

Paper IV independently declares its exact nine-point arrangement, Rubik registration, and conditional interpretation. CCS v2 is not a certificate or premise for that paper.

Paper V comparison. Paper III proves that support-graph reachability need not survive projected matrix composition. This motivates, but does not prove, the later minimal-data problem for general sectorized observable frameworks:

$$R_1(i,j;g) = 1 \iff Q_i X_g Q_j \neq 0, \qquad R_2(i,j;g,h) = 1 \iff Q_i [X_g, X_h] Q_j \neq 0.$$

The CCS does not define or certify Paper V's typed word, commutator, or Lie objects. In particular, it makes no claim that (R_1, R_2) universally determines first accessibility depth.

II.3 Historical Generator-Family Census

Computational observation. This finite census compares four declared generator families. It does not establish a universality class or an exact arithmetic-field classification.

Purpose. This census compares how registered spectral and direct-support quantities change across four explicitly declared face-turn families.

Dependencies. `CubieSpectralOperator.from_gens_dict`, `center_decomposition`, `transport_graph`, `build_per_axis_ops`, `full_commutant_combinatorial`.

Outputs. Four-family numerical comparison table and candidate regularities.

II.3.1 Four Generator Families

Family	|S|	Description
18-full	18	All face turns: {R,R',R2, U,U',U2, F,F',F2, L,L',L2, D,D',D2, B,B',B2}
12-quarter	12	Quarter-turns only: {R,R', U,U', F,F', L,L', D,D', B,B'}
6-half	6	Half-turns only: {R2, U2, F2, L2, D2, B2}
9-face-pos	9	Single-side subset: {F/B/R/U faces only, axis side = +1, quarter-turns + half-turns} (legacy label: "12-face")

Note: The "9-face-pos" family is the specific 9-generator subset historically labelled "12-face" in earlier CCS revisions. The numeric prefix matches the actual generator count; the legacy label is retained in cross-references where otherwise noted. Exact composition: F/B/R/U faces, axis side = +1, quarter-turns + half-turns, 9 generators.

II.3.2 Invariant Table

Invariant	18-full	12-quarter	6-half	9-face-pos	Stable?
Layer count	6	6	3	18	**NO**
Rational spectrum	True	True	True	False	**NO**
Layer dimensions	see below	see below	see below	see below	**NO**
Σ dim = 228	True	True	True	True	**YES**
Directed off-diagonal blocks	20	20	20	—	**Observed equal**
Canonical graph form	Sparse, non-star	Not classified here	Not classified here	—	**Not assessed**
Cross-block K	0	0	0	—	**YES**
Legacy algebra-dimension diagnostic	2	2	2	2	**Historical only**
$\|[\mathbf{Q}T^0,\mathbf{Q}T^1]\|$	2.915	N/A	N/A	7.591	**NO**
EP fraction	93.9%	N/A	N/A	72.2%	**NO**
EP block dim	144	144	144	144	**YES**
k=5 vacancy	True	True	N/A	N/A	**YES** (where applicable)

Layer dimensions per family:

Family	Layers	Dimensions
18-full	6	[20, 2, 39, 26, 106, 35]
12-quarter	6	[20, 41, 66, 65, 28, 8]
6-half	3	[57, 78, 93]
9-face-pos	18	[68, 2, 1, …]

II.3.3 Invariant Classification

Representation-determined records:

1. **Total dimension** = 228 — the group representation is the same object regardless of generator subset.
2. **EP block dimension** = 144 — the block structure of ρ is generator-independent.
3. **Cross-block K = 0** — direct blocks between disjoint physical carrier blocks vanish because every $\rho(g)$ and every declared sector projector is block diagonal.

Generator-family-conditioned observations:

1. **Layer count** — varies from 3 (6-half, fully commutative) to 18 (9-face-pos, symmetry-broken). The 18-full and 12-quarter both yield 6 layers — quarter-turn completeness across all faces is the minimal condition for full spectral resolution.
2. **Rational spectrum** — requires face-symmetric generator sets. Breaking face symmetry (9-face-pos) introduces irrational eigenvalues. The half-turn family preserves rationality but collapses the spectrum to 3 layers.
3. **Noncommutativity magnitude** — 9-face-pos has $\|[QT^0,QT^1]\| = 7.59$ vs 2.91 for 18-full. EP fraction drops from 93.9% to 72.2% — noncommutativity distributes more evenly across blocks.

II.3.4 Scope and Relation to §II.4

This section studies four generator families at fixed points. Section II.4 records a larger finite coverage sequence and numerical recognition against $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{5})$. Neither table is an exhaustive classification of generator subsets.

II.3.5 Computational Details

- Archived source: `experiments/paper3/archive/persistence_bridge.py`, Phase B
- Method: Build `CubieSpectralOperator.from_gens_dict(gens)` for each family, call `center_decomposition()`, `transport_graph()`, `build_per_axis_ops()`, `full_commutant_combinatorial()`.
- QT^0/QT^1 noncommutativity only computable when both quarter-turn and half-turn axes are present (requires axis-0 QT0/QT1 decomposition); N/A for families without half-turns or without both axis directions.
- Commutant computed via index-pair orbit decomposition (BFS on (i,j) → (π_g(i), π_g(j))).
- All computations use TOL = 10^{-10}, seed = 42.

II.4 Registered Generator-Family Arithmetic Contrast

Computational observation. The displayed fields are numerical recognitions against explicit candidate expressions. This archive does not provide exact characteristic or minimal-polynomial certificates for every family and does not claim a mathematical phase transition.

Purpose. Record spectral changes along one declared sequence of generator subsets as the number of retained face turns decreases.

Dependencies. `CubieSpectralOperator.lite`, `helpers.is_rational_form`, `helpers.is_in_qsqrt5`, `BLOCK_RANGES`.

Outputs. Generator coverage continuum table, eigenvalue bifurcation track, phase boundary characterization, block-level irrationality localization, transport topology deformation.

II.4.1 Generator Coverage Continuum

Construction. Start with all 18 generators. Remove generators in pairs (face + anti-face to preserve algebraic balance), stepping: $18 \to 16 \to 12 \to 10 \to 8 \to 6$

Family	|S|	Layers	All Q?	Field	5/9 dim	2/3 dim	Notes
n=18	18	6	True	Q	106	26	Canonical, full face-turn group
n=16	16	9	False	Q(√5)	26	0	Drop axis-2 (F/B) half-turns
n=12	12	8	True	Q	0	66	Quarter-turns only
n=10	10	5	True	Q	0	0	Quarter-turns minus axis-2
n=8	8	7	False	Q(√5)	0	0	Axes 0 and 2 only, no half-turns
n=6	6	3	True	Q	0	78	Half-turns only

Registered contrast. Values numerically matching $\mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}$ occur at the $n = 16$ and $n = 8$ points of this declared sequence. Although this finite pattern motivates an exact arithmetic problem, it does not show that incomplete face coverage is either necessary or sufficient for a field extension.

The corresponding first-version phase-transition visualization is retained only as repository provenance. The table above is the current archive record.

II.4.2 Eigenvalue Bifurcation Data

Table — Eigenvalue Spectrum per Generator Family.

Family	|S|	Eigenvalues	Field	Irrational values
n=18	18	$\{1, 8/9, 7/9, 2/3, 5/9, 1/3\}$	Q	—
n=16	16	$\{1, 7/8, 0.827, 3/4, 5/8, 0.548, 1/2, 3/8, 1/4\}$	Q(√5)	$(11 \pm \sqrt{5})/16$
n=12	12	$\{1, 5/6, 2/3, 2/3, 1/2, 1/3, 1/3, 0\}$	Q	—
n=10	10	$\{1, 4/5, 3/5, 2/5, 1/5\}$	Q	—
n=8	8	$\{1, 0.905, 3/4, 1/2, 0.345, 1/4, 0\}$	Q(√5)	$(5 \pm \sqrt{5})/8$
n=6	6	$\{1, 2/3, 1/3\}$	Q	—

Block-level irrationality localization (n=8). Irrational eigenvalues are confined to non-commutative blocks: EP block (primary carrier), EO block (mirrors EP). CP block: all eigenvalues rational (Q_3 Hamming scheme's Krawtchouk eigenvalues are generator-subset-stable). CO block: all eigenvalues rational.

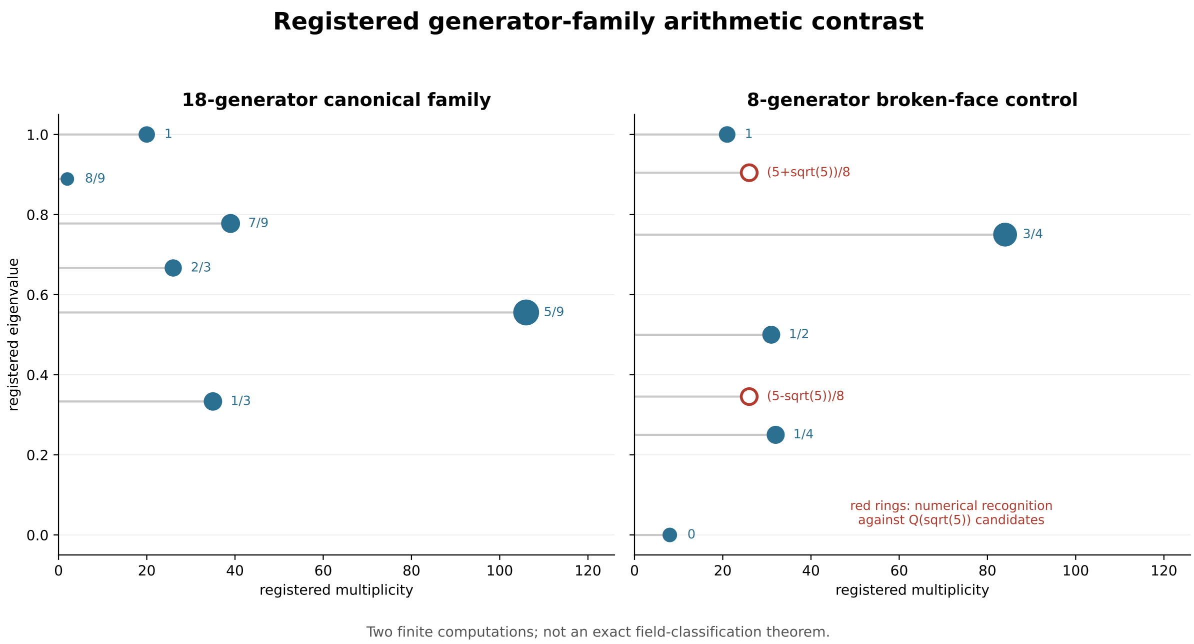
II.4.3 Transport Topology Deformation

Family	Sectors	K edges	Hub (deg)	Cross-block K	Topology class
n=18	9	20 directed / 10 unordered	S6 (5)	0	Sparse, non-star
n=16	—	—	—	—	Dense (irrational splitting expands sector count)
n=12	—	—	—	—	Collapsed (degeneracy absorbs 5/9 layer)
n=10	—	—	—	—	Sparse (fewer layers → fewer possible edges)
n=8	7	28	S6 (5)	0	Hyper-connected (irrational intruders create more edges)
n=6	3	3	S2 (2)	0	Minimal (commutative limit, complete graph K_3)

Observed cross-block persistence. The cross-block transport prohibition (K_$\alpha\beta$ = 0 for α,β in disjoint blocks) holds at every n verified. This follows from ρ(g) being block-diagonal — a property of the representation, independent of which generators are selected.

Observed hub recurrence. A degree-five registered hub also appears at the $n = 8$ point. No theorem identifies hub status with $\mathrm{Supp}_{\mathrm{nc}}$ intersection or guarantees persistence under generator variation.

II.4.4 Bounded Arithmetic Contrast



CCS Figure C20. Current source-addressed comparison of the canonical family and the eight-generator broken-face control. Red rings denote numerical recognition against displayed quadratic candidates.

Observed pattern. The two highlighted values in the broken-face control are numerically recognized against $\mathbb{Q}(\sqrt{5})$ candidates. The figure does not provide an exact field certificate or classify all generator subsets.

II.4.5 Computational Details

- Archived source: `experiments/paper3/archive/persistence_bridge.py`, Phase C
- Method: For each n = 18, 16, 12, 10, 8, 6, select a generator subset of size n from the 18 face-turn moves, build A = (1/n) Σ ρ(s), diagonalize.
- n=18: full 18 generators
- n=16: all moves except F2 and B2 (axis-2 half-turns)
- n=12: quarter-turns only (direction=$\pm$1, all 6 faces)
- n=10: quarter-turns without axis-2 faces (remove F, F', B, B')
- n=8: quarter-turns only, axes 0 and 2 (R,R',L,L',F,F',B,B')
- n=6: half-turns only (direction=2, all 6 faces)
- Irrationality detection: `helpers.is_rational_form(lam, 18)` and `helpers.is_in_qsqrt5(lam)`.
- Block-level decomposition: restrict A to block submatrices and diagonalize.
- All computations use TOL = 10^{-10}.

---

6 Historical Derivation Index

The 2026-07-26 mother source contained a compact invariant hierarchy followed by complete derivation chapters for the former combined-paper narrative. Those chapters remain below in the Markdown source for versioned provenance, but are excluded from the CCS v2 PDF. Current theorem statements and proofs belong to the independently maintained Papers I–III; the archive must not duplicate or override them.

Appendix A Computational Stability

A.1 Tolerance Regime

The canonical tolerance regime is:

Symbol	Value	Scope
`TOL`	10^{-10}	Numerical equality assertions
`TOL_K`	0.05	Transport edge detection threshold
`TOL_KAPPA`	10^{-6}	Historical κ-array sanity floor (logm noise ceiling)
`SPECTRAL_DECIMALS`	6	Canonical eigenvalue key rounding
`CENTER_CLUSTER_TOL`	10^{-8}	Sector merge clustering
`tol`	10^{-6}	Default operator tolerance

The archived realization uses these tolerance values. A computation using a different `SPECTRAL_DECIMALS`, `CENTER_CLUSTER_TOL`, or `TOL_K` is a distinct registered realization and should report a comparison sweep.

The registered sector decomposition uses `CENTER_CLUSTER_TOL` $= 10^{-8}$. A different clustering tolerance should be identified explicitly and cross-validated against the Part I tables.

Historical numerical robustness. The recorded layer, sector, edge, and graph-only candidate counts remain invariant under the tested tolerances and seeds. The first-version audit is archived at `experiments/paper3/archive/stability_sweep.py` and does not certify matrix composition.

Graph-candidate threshold stability. The five graph-only candidate pairs are invariant under the recorded threshold sweep. The archived script is `experiments/paper3/archive/t7_threshold_sensitivity.py`; this statement does not promote graph reachability to operator composition.

A.2 Norm and Projector Conventions

Unless a record states otherwise, archived matrix norms are Frobenius: $\|X\|_F = \sqrt{\sum |x_{ij}|^2}$.

Archived projectors use $P = VV^H$, where V has orthonormal columns from `numpy.linalg.eigh`, so $\mathrm{Tr}(P) = \dim(\text{subspace})$.

The registered generator weighting is the uniform average $A = \frac{1}{|S|}\sum_{s\in S}\rho(s)$.

Where a seed is used, the registered value is `np.random.seed(42)`. Deterministic audits should not depend on random state.

A.3 Claim-Status Routing

Numerical stability and mathematical claim status are separate. The current repository uses four paper claim levels; CCS v2 adds a history tag solely for archive routing.

Status	Use in CCS v2
Theorem / exact derivation	Restate only with explicit hypotheses and proof; cite the owning independent paper when used externally.
Computational Certificate	Record realization, dtype, tolerance, registration, algorithm, artifact, and reproducible script.
Computational Observation	Record a finite pattern, contrast, or numerical recognition without promotion beyond the tested realization.
Research Program	Record conjectures, proposed hierarchies, genericity questions, and future experiments.
Historical / Withdrawn	Preserve correction provenance without treating the item as a current claim.

Passing recomputation, basis, permutation, or tolerance checks strengthens a computational record but does not by itself change its status. In particular, the candidate ambient-commutant dimension, first-version κ/T7 diagnostics, and moving accessibility hierarchies are not promoted by this appendix.

A.4 Failure Modes

All observed failure modes fall into four categories:

1. **Spectral degeneracy artifacts** — eigensolver splitting, eigenvector mixing

2. **Finite-precision linear algebra instability** — SVD thresholding, null-space drift
3. **Representation-construction defects** — pre-ρ-fix orientation sign inconsistency
4. **Algorithmic non-canonicality** — generator ordering dependence, randomized under-convergence

The canonical r2 pipeline eliminates categories (3) and (4) by construction. Categories (1) and (2) are controlled via explicit tolerance engineering.

A.4.1 Pre-ρ-fix Representation Defect (Category 3)

The most consequential recorded implementation failure was an inconsistent sign convention in the EP orientation sub-block. It caused $\rho(g)\rho(h) \neq \rho(gh)$, thereby violating the homomorphism property. This erased the $V_{8/9}$ layer ($k = 1$, 2-dim) entirely — it was absorbed into adjacent layers by numerical accident.

> **Warning.** Pre-fix data archived and must not be cited. Representation defects differ qualitatively from numerical issues — they propagate into structural claims, not just numerical values. Block-wise homomorphism verification is mandatory. **Status**: Resolved in r2.

A.4.2 Legacy Error Catalog (Categories 1, 2, 4)

The private raw archive retains the full historical logs; the public error catalog is:

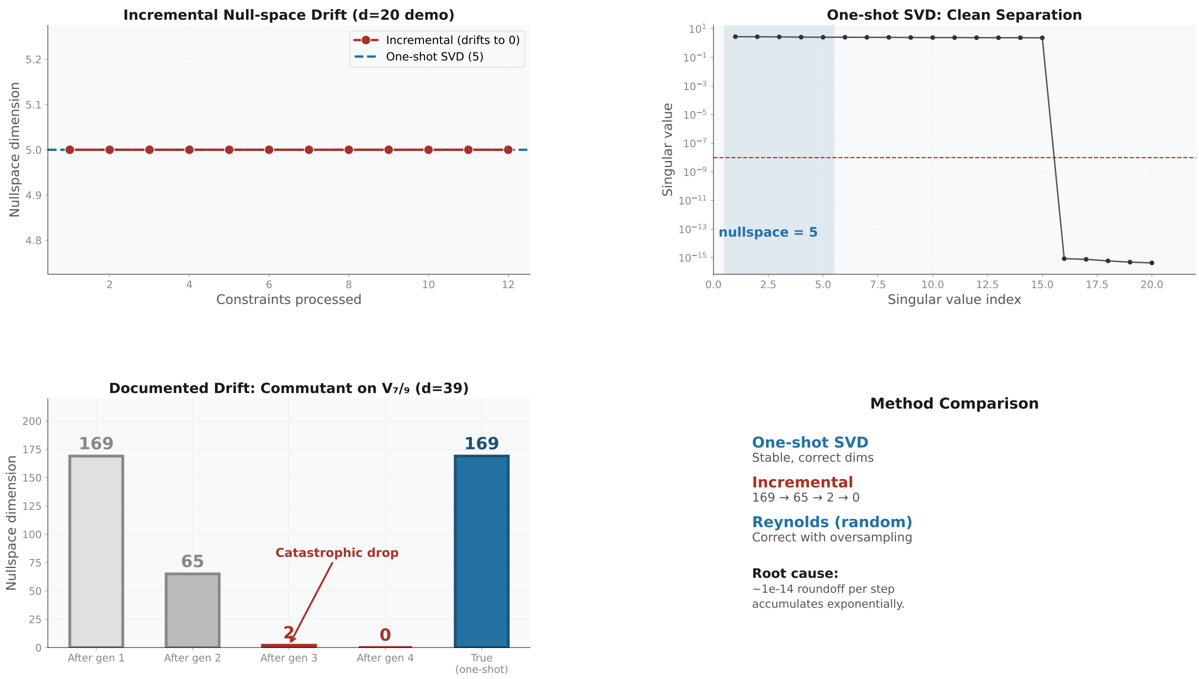
- E.1 Eigensolver accidental splitting ($\rightarrow$ 11 raw sectors merged to 9)
- E.2 SVD rank threshold instability ($\rightarrow$ scale-invariant threshold + one-shot SVD)
- E.3 Near-degenerate eigenvalue mixing (does not occur — minimum gap $1/9 \approx 0.111$)
- E.4 Generator ordering artifacts (mitigated by permutation-invariant averaging)
- E.5 Randomized Reynolds under-convergence (adequate for current $d \leq 106$)
- E.6 Sector permutation across recomputation (does not occur — well-separated triples)
- E.7 Lie generator non-Hermiticity drift (mitigated by explicit Hermitianization)
- E.8 Incremental null-space drift (design rejected — one-shot SVD used instead)

A.5 Numerical Registration and Stability Policy

The working principle of the archived numerical pipeline is:

> **Registration policy.** Register first, analyze second.

Numerically equivalent realizations must be reduced to a declared registration before their tables are compared. The following checks support the stability of a finite record, but passing them does not automatically promote an observation or certificate to the status of a theorem:



CCS Figure C5. Null-space drift diagnostics: incremental intersection vs one-shot SVD.

1. **Recomputation** — same code, same parameters → same result to within prescribed tolerance.
2. **Generator permutation** — invariant under $S_6 \times \mathbb{Z}_2$ face relabeling.
3. **Basis changes** — invariant under $U(n)$ gauge freedom inside degenerate eigenspaces.
4. **Tolerance perturbation** — stable under perturbation of any tolerance within the prescribed regime (§A.1).

Records failing a relevant check remain computational observations or research program items. Current claim status is assigned under the four-level contract in the independent paper, not by this appendix.

A.6 Gauge Freedoms and Registered Choices

The archived implementation fixes the following representation choices. Alternative choices are admissible when they are declared and cross-validated.

#	Freedom	Canonical fixing
D.1	Eigenspace basis ($U(d_\lambda)$)	`numpy.linalg.eigh` — first nonzero element positive. Real eigenvectors where possible (real symmetric matrix).
D.2	Sector label ordering	**CCS canonical**: sort by $k = 9(1 - \lambda_{18})$ ascending; within fixed k, sort by dimension ascending. Labels S1–S9 are frozen by Table C3 (§1.4). All figures, tables, and transport tensors MUST use this ordering.
D.3	Sector merging (accidental split)	Merge sectors whose (λ_{18}, λ_{QT}, λ_{HT}) triples differ by $< 10^{-8}$ in all three coordinates. The 11→9 merge pattern is deterministic.
D.4	Generator labels ($S_6 \times \mathbb{Z}_2$)	`CubieMove.prim_moves` enumeration order. Spectral identity (layers, dimensions, projectors) is label-invariant.
D.5	Isotypic multiplicity ($\mathrm{GL}(m, \mathbb{C})$)	Commutant basis from orbit-enumeration construction. Gram-Schmidt orthogonalized.
D.6	Layer key representation	`SPECTRAL_DECIMALS = 6`. Canonical keys: $\lambda = 1 - k/9$, $k \in \{0, 1, 2, 3, 4, 6\} \to [1, 8/9, 7/9, 2/3, 5/9, 1/3]$.
D.7	Ambient-commutant candidate basis	Gram-Schmidt orthogonalized conjugacy-class orbit sums. The registered numerical candidate dimension is 610; an exact certificate remains open.

Projectors, transport strengths, and layer dimensions should agree under valid changes of basis up to the declared matching and numerical tolerances. First-version κ_d diagnostics and the candidate ambient-commutant dimension are retained as archived computational records, not theorem-level gauge invariants established by CCS v2.

Appendix B Provenance

Purpose. Record traceable lineage from experiment scripts to structured artifacts, archive tables, figures, and paper-level computational statements.

Scope. Data flow diagram, artifact/experiment provenance map, paper usage map, and figure mapping.

Dependencies. The paper-local validation and result records indexed below.

Outputs. A reviewable provenance map linking retained numerical records to their producing experiments and, where applicable, their consuming papers.

This part indexes provenance. It does not assign mathematical claim status.

B.1 Data Flow

```
experiments and validation scripts -> structured artifacts -> papers
                                  \-> generated figures
                                  \-> CCS v2 review index
private raw archive               -> provenance only
```

Executable scripts and structured outputs form the certificate layer. CCS v2 indexes selected values and observations for human review. The private raw archive retains failed experiments and older revisions as provenance only.

B.2 Artifact and Experiment Provenance Map

Claim	Primary experiment
6-layer spectrum, dims, block support	`experiments/paper1/validation/spectral_ladder.py`
registered $k = 5$ vacancy	`experiments/paper1/validation/k_absence.py`
Projector algebra ($P_iP_j = \delta_{ij}P_i$)	`experiments/paper1/validation/projector_algebra.py`
9 QT/HT joint-spectral sectors	`experiments/paper2/validation/primitive_sectors.py`
K matrix, transport graph	`experiments/paper2/validation/transport_graph.py`
Block noncommutativity	`experiments/paper2/validation/supp_nc.py`
EP algebra census	`experiments/paper2/validation/ep_algebra.py`
Graph/operator separation for five canonical paths	`experiments/paper3/validation/composition_obstruction.py`
Withdrawn spectral-layer π map	`experiments/paper2/archive/commutant_pi_map.py` (provenance only)
Withdrawn compressed-commutant census	`experiments/paper1/archive/isotypic_decomposition.py` (provenance only)

B.3 Subject Navigation

Subject	Primary CCS reference	Archived material
Averaging-operator spectrum	§§1.1–1.8	Layers, block reductions, and extended census tables
Direct transport	§§1.4, 2.1–2.2, 2.7–2.8	Sectors, K matrix, localizer data, and EP algebra
Projected composition	§2.5	Optional copy of the five matrix-obstruction records

B.4 Figure Boundary

Figures used by the current archive are placed beside the records they illustrate and are explained by their captions. Historical and withdrawn images remain repository provenance; they are not indexed in this release and do not enlarge the claims of the independent papers.

Appendix C Figure Provenance

The release PDF includes only figures placed directly in the relevant archive sections. Historical images and presentation-build details are maintained in the repository rather than repeated here.

Appendix D Implementation Notes

Purpose. Document computational methods that underlie the numerical values in Parts I–II. These certify reproducibility without interrupting the mathematical narrative of the main papers.

Scope. Representation construction, projector computation, transport and Lie generator algorithms, commutant computation methods, computational complexity table.

Dependencies. Part 0.5 (canonical API), Part I (canonical objects), `rime/` source modules.

Outputs. Complete algorithmic specification sufficient for independent reimplementation.

This part provides the algorithmic specification sufficient for independent reimplementation of the canonical computation.

Computational methods that underlie the numerical values in Parts I–II. These belong in the supplement, not in the main papers — they certify reproducibility without interrupting the mathematical narrative.

D.1 Representation Construction

$\rho : G \to \mathrm{GL}(228, \mathbb{C})$ is built as a permutation+phase representation on corner and edge cubie states. Corner positions are indexed by sign vectors $\{x \in \{\pm 1\}^3\}$; edge positions by vectors in $\{x \in \{\pm 1, 0\}^3 : \sum_i |x_i| = 2\}$.

On the **permutation blocks** (CP, EP), generators act by permutation matrices: $\rho(g)_{ij} = 1$ if position j maps to position i, 0 otherwise. These are integer matrices: $\rho_{\mathrm{cp}}(g) \in M_{64}(\mathbb{Z})$, $\rho_{\mathrm{ep}}(g) \in M_{144}(\mathbb{Z})$.

On the **orientation blocks** (CO, EO), generators additionally multiply by a phase factor on each affected index: $\rho(g)_{ij} \in \{0, \pm 1\}$ (EO, $\mathbb{Z}_2$) or $\rho(g)_{ij} \in \{0, 1, \omega, \omega^2\}$ (CO, $\mathbb{Z}_3$, with $\omega = e^{2\pi i/3}$).

Post-ρ-fix invariant: $\|\rho(g)\rho(h) - \rho(gh)\|_F < 3 \times 10^{-8}$ on all blocks — the homomorphism property is exact to machine precision. Verified on 15 random products.

D.2 Projector Computation

Spectral projectors P_λ are computed via `numpy.linalg.eigh` on A_{18}:

$$P_\lambda = V_\lambda V_\lambda^H$$

where $V_\lambda \in \mathbb{C}^{228 \times d_\lambda}$ has orthonormal columns (Schatten normalization). $\mathrm{Tr}(P_\lambda) = d_\lambda$, $P_\lambda^2 = P_\lambda$, $P_\lambda P_\mu = \delta_{\lambda\mu} P_\lambda$.

Sector projectors P_{S_k} are obtained by numerical joint diagonalization of A_{18}, $\mathrm{QT}_{\mathrm{all}}$, and $\mathrm{HT}_{\mathrm{all}}$ after the declared commutator audit. Their registered joint clusters group into nine sectors at CENTER_CLUSTER_TOL $= 10^{-8}$. Exact simultaneous spectral projectors require exact commuting-Hermitian registration.

D.3 Transport and Lie Generators

Transport: $K_{\alpha\beta} = \max_{g \in S} \|P_\alpha \rho(g) P_\beta\|_F$ — enumerates all 18 generators, no optimization needed.

Historical principal-log registration: $A_g = \mathrm{logm}(\rho(g))$ via `scipy.linalg.logm`, using the declared numerical branch. The embedding residual is $\max_g \|\exp(A_g) - \rho(g)\| < 3 \times 10^{-15}$. These arrays are retained for first-version reproducibility and are not the anti-Hermitian-part family or a current exact Lie-depth certificate.

Historical commutator arrays: κ_1 uses $[A_g, A_h]$ for all 153 unordered generator pairs. The partial κ_2 enumeration is exploratory and does not certify Lie closure or saturation.

D.4 Commutant Computation

Ambient commutant candidate (registered dimension 610): a combinatorial orbit-enumeration implementation followed by numerical orthogonalization. The current archive does not contain an independent exact rank/nullity certificate, so 610 remains an unpromoted candidate.

Per-layer commutant:

- $d_\lambda \le 50$: One-shot SVD on the Kronecker constraint matrix after linear dependency reduction. The constraint is $CX = 0$ where C stacks $G^T \otimes I - I \otimes G$ for each independent generator. Rank threshold: $\mathrm{tol} \cdot \max(1.0, s_0) \cdot \max(C.\mathrm{shape})$.
- $d_\lambda > 50$ (only $V_{5/9}$, $d = 106$): Randomized Reynolds iteration — sample budget $\min(6d, 250)$, 8 iterations of exact projection, convergence to machine precision.

D.5 Computational Cost

$N = 228$, $d_\lambda \le 106$, $|S| = 18$, $K = 3$ (Center operators).

Operation	Complexity	Wall time
A_{18} eigendecomposition	$O(d^3)$	< 1s
center_decomposition()	$O(K \cdot d^3)$	< 5s
Ambient commutant candidate	$O(d^2 \cdot \lvert\mathrm{Conj}(G)\rvert)$	< 1s
Layer commutant ($d_\lambda \le 50$)	$O(d_\lambda^6)$	~1s each
Layer commutant ($d_\lambda = 106$)	$O(d_\lambda^3 \cdot N_{\mathrm{iter}})$	~30s
transport_kappa()	$O(\lvert S\rvert \cdot d^3)$	~2s
kappa_depth(2)	$O(\binom{153}{2} \cdot d^3)$	~10s
π map SVD (610×966)	$O(610 \cdot 966 \cdot \min(610, 966))$	< 1s

Total wall time for full canonical recomputation: ~5–10 minutes on commodity hardware.

Appendix E Archive Status Register

Purpose. Route retained CCS v2 material to its current status without restating the independent papers' theorem spines.

Scope. Selected Paper I–II data records, archived extensions, open questions, and withdrawn first-version interpretations.

Dependencies. The current Paper I and Paper II ‘`Claim Status and Boundary'' sections, declared executable artifacts, and`HISTORY.md‘.

Outputs. A compact routing table. It is not a theorem register.

The independent papers control theorem and certificate wording. CCS v2 retains extended human-readable records only.

Archive material	Current status	Controlling source
Six-layer A_{18} census and block records	Mixed exact derivation, computational certificate, and observation	Paper I claim-status section and Paper I scripts
Trace-rationality and partition-integrality statements	Theorem under the hypotheses stated in Paper I; no canonical-face application is asserted here	Paper I
Nine QH sectors, ten direct edges, and EP algebra census	Computational certificate	Paper II and Paper II scripts
Generator-family arithmetic and transport extensions	Computational observation	CCS v2 tables and archived scripts
Candidate ambient-commutant dimension 610	Research program / unpromoted numerical candidate	CCS v2 provenance
Graph/operator composition obstruction	Outside CCS authority; independently certified	Paper III and its matrix audit
C0, T7 morphisms, strict word/Lie containment, and old κ completion narrative	Historical / withdrawn	`HISTORY.md`
Moving spectral/accessibility hierarchies	Research program unless separately certified in the owning paper	Papers IV–VII claim boundaries

The repository may retain older tables below in source history, but they are excluded from the current CCS output because their numbering and promotion language predate the v2 paper revisions.

Appendix F Future Directions and Verification Scope

Purpose. Collect CCS-adjacent open problems and computational boundaries. Nothing here is claimed as proven or imported into revised Paper III.

Verification status. The current records include the mixed-status six-layer census, nine numerically registered QH sectors, the direct transport matrix, sector non-invariance residuals, and five graph/operator composition obstructions in the 228-dimensional Rubik realization. They do not establish the former T7 strict-containment theorem, a layerwise G-isotypic decomposition, or an exact 610-dimensional ambient-commutant theorem. Exact statements and finite numerical certificates are separated in Papers I–III.

F.1 Structural Generalization

Broader finite-group representations. CCS experiments include the Rubik cube (228-dim), S_3 nat$\oplus$reg (9-dim), and S_3 reg$\oplus$reg (12-dim). It remains open whether the Paper II transport architecture generalizes to non-block-diagonal representations, non-symmetric generator families, or infinite discrete groups. The central structural question is which features are G-determined and which are generator-conditioned. Whether $\mathrm{Supp}_{\mathrm{nc}}$ extends canonically to non-permutation or non-semisimple transport geometries is unknown.

Alternative spectral algebras. The registered Rubik algebra $\mathcal{B}_{\mathrm{QH}}$ produces nine sectors with a sparse direct transport graph. Alternative commuting algebras may change both the sector count and the support/composition relation. No refinement-invariance theorem for a compositional gap is currently claimed.

Graph-to-composition promotion. The canonical audit instead establishes that a two-step support path need not survive matrix multiplication. A current open problem is to identify transversality, rank, or image–kernel conditions under which graph reachability does imply operator reachability. Comparisons with Lie-generated accessibility require a separate certificate after this operator-level relation is fixed.

Further structural questions. Whether the Type I/II classification, the M_2 overlap pattern, or graph/composition obstructions appear in non-permutation representations is open. The candidate ambient-commutant dimension must first receive an independent exact certificate before it is related to transport-graph statistics.

F.2 Accessibility Questions

Algebraic characterization of noncommutative support. Paper II defines $\mathrm{Supp}_{\mathrm{nc}}(\alpha)$ from the family-level maximum over the three per-axis QT commutator pairs. In the registered Rubik realization, shared support selects 15 unordered candidates: nine Type I labelled direct edges and six nonedges. It is therefore a localizer, not an edge characterization. Whether a sharper support notion can be derived algebraically from the represented finite-dimensional $*$-algebra remains open.

Promotion conditions. The former C0–C3 characterization is withdrawn. Candidate replacement hypotheses should act directly on the projected factors, for example through image–kernel transversality, rank protection, singular-value lower bounds, or compatible block support.

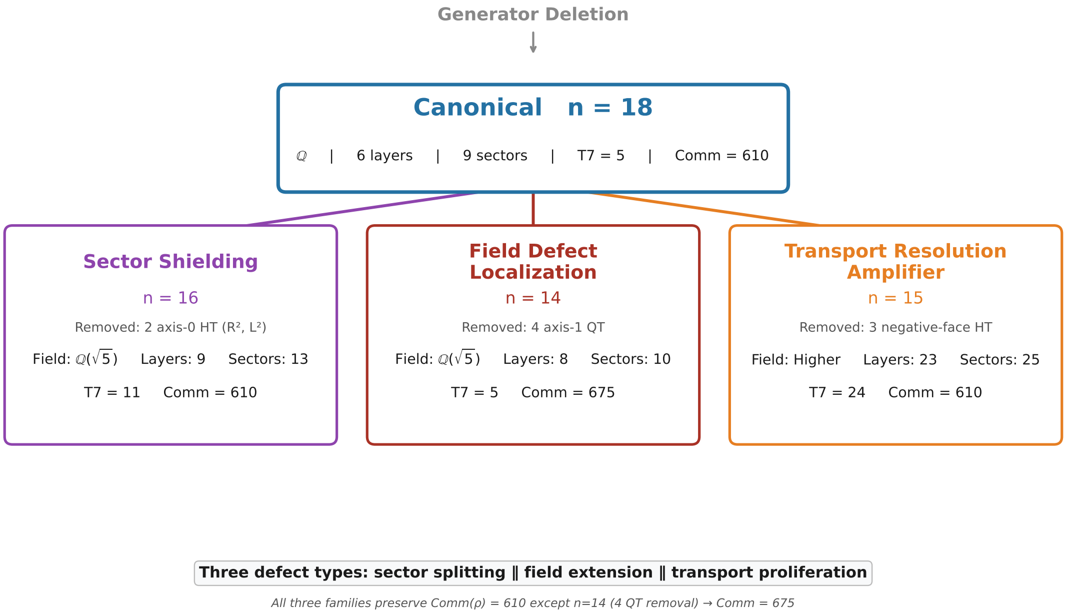
Classification of composition obstructions. The current five witnesses are explained by physical-block image–kernel mismatch. Whether other systems exhibit incidence, cancellation, or rank-loss obstructions without block separation is open.

F.3 Algebraic Extensions

Generalized transport algebras. The direct-support norm $K_{\alpha\beta} = \max_g \|Q_\alpha \rho(g) Q_\beta\|_F$ is defined for the declared sectorization and operator family. What changes when the projectors come from another certified commuting-normal registration? Routed products, full words, commutators, and Lie depth must remain separate typed objects. The first-version κ arrays do not provide the missing bridge.

Refinement questions beyond M_2. The EP census exhibits four registered M_2 components, while the direct graph and localizer record a finite overlap pattern. No theorem states that the presence of M_2 alone forbids every transport-compatible refinement. Identifying the additional hypotheses needed for a refinement obstruction remains open.

Historical Generator Defect Taxonomy. Four generator families were constructed by selective deletion from the 18-generator canonical set. The first-version script is archived at `experiments/paper3/archive/generator_defect_taxonomy.py` and is not current matrix-composition evidence.



CCS Figure C18. Generator defect taxonomy: canonical n=18 and three defect families — Sector Shielding (n=16), Field Defect Localization (n=14), Transport Resolution Amplifier (n=15).

F.4 Computational Directions

Scalable commutant extraction. The current commutant computation uses generator reduction + one-shot SVD ($d \leq 50$) or randomized Reynolds ($d > 50$). For representations beyond ~1000 dimensions, both methods become impractical. A scalable commutant algorithm — perhaps exploiting sparse generator structure or block-diagonal preconditioning — is open.

Automated typed audit. A future pipeline may take a declared representation, sectorization, operator family, dtype, and tolerance policy and emit separate direct-support, routed-product, full-word, commutator, and Lie-closure records. It must preserve these types rather than reconstruct the withdrawn $K \to \kappa_0 \to \kappa_1 \to$ T7 ladder.

Exact QH-algebra reconstruction. The commuting algebra $\mathcal{B}_{\mathrm{QH}} = \mathrm{alg}(A, \mathrm{QT}_{\mathrm{all}}, \mathrm{HT}_{\mathrm{all}})$ is presently supported by numerical commutation and joint-sector certificates. An exact reconstruction would promote this numerical object without identifying it with the center or the full ambient commutant.

Separate Lie-accessibility audit. First-version κ_d calculations are archived diagnostics. Any future comparison between Lie accessibility and projected word composition requires a new claim-specific certificate.

F.5 Structural Scope Boundary

What the current CCS does NOT claim:

Claim	Status
A support-graph path implies nonzero projected composition	**Refuted in general.** The five canonical paths are obstruction witnesses.
The CCS applies to AGI, cognition, planning, solver algorithms, robotics	**Not claimed.** Its scope is finite-group representation computation.
The CCS is a general classification of finite-group representations	**Not claimed.** Rubik is one finite computational realization.
The first-version κ_d/T7 hierarchy proves a composition gap	**Withdrawn.** A separate operator-level certificate is required.
The registered direct graph is directed or asymmetric	**Not supported.** The inverse-closed family gives a symmetric K matrix to the declared numerical tolerance.

Code availability. Code and computational certificates are available in the RIME repository.

End of the CCS v2 computational companion archive. Independent papers and declared executable artifacts control current claims and certificates; withdrawn source sections are retained for provenance only.